

ADAPT-Tracker: Adaptive dynamic perception and trajectory-consistent network for aerial multi-object tracking[☆]

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ABSTRACT

Aerial Multi-object tracking is hindered by coupled dynamics in which rapid UAV ego-motion disrupts inter-frame alignment and stochastic target motion amplifies non-linear observation noise, jointly breaking temporal consistency—especially for tiny objects. Existing trackers struggle in this compound regime because convolutional backbones suppress small-target cues under clutter, while constant-velocity motion models are brittle to maneuver-induced jitter and intermittent detections. We propose ADAPT-Tracker, a unified framework that couples scale-aware representation learning with high-order geometric trajectory refinement. On the perception side, Motion-Aware Receptive Field Modulation (MRFM) decomposes features into complementary granularity components to preserve fine-grained target details while suppressing distractors, and performs structure-preserving channel mixing to strengthen small-object representations. On the motion side, High-Order Geometric Stabilized Trajectory Refinement (H-GSTR) conducts bi-directional trajectory inpainting to restore continuity under occlusion and missed detections, and applies curvature-regularized kinematics to stabilize maneuver-induced jitter. Beyond overall metrics, a sequence-level recovery study on UAVDT shows that H-GSTR recovers 8.66% additional valid detections on average, peaking at 27.27% in high-viewpoint small-target scenes. Extensive experiments on UAVDT and VisDrone-MOT demonstrate robustness under scale variation and abrupt motion, improving over the strongest competing method by approximately +2.00 (HOTA), +3.00 (MOTA), and +1.50 (IDF1) percentage points on VisDrone-MOT. Consistent superiority is observed on the UAVDT benchmark.

1. Introduction

Unmanned Aerial Vehicles (UAVs) have become indispensable platforms for intelligent Earth observation and remote sensing, enabling a wide range of applications such as wide-area surveillance, traffic analysis, change detection, and disaster response (L. Jiang et al., 2025; Zou et al., 2025; Y. Zhang et al., 2025; Jing et al., 2025). Compared with static surveillance cameras or ego-centric driving setups, UAV-based multi-object tracking (MOT) faces a uniquely coupled dynamic environment: the unconstrained 6-DoF camera motion continuously changes the viewing geometry, while ground targets exhibit stochastic and diverse kinematics. This coupling gives rise to a *dual-dynamic dilemma* in aerial tracking: (i) perception instability, where abrupt scale swings, viewpoint distortion, and motion blur suppress tiny-object features and destabilize appearance cues; and (ii) trajectory inconsistency, where ego-motion and target motion are entangled, breaking

classical motion assumptions and causing frequent identity switches and fragmented trajectories. As observed in recent UAV trackers such as ArbiTrack (Chen et al., 2025a) and DroneMOT (P. Wang et al., 2024), aggressive maneuvers (e.g., rapid altitude changes, pitching, and yawing) can trigger drastic scale variation and non-linear perspective deformation, under which trackers effective on ground-level videos often fail to maintain both reliable target perception and geometrically consistent trajectories (Xie et al., 2025; Simões et al., 2026).

Existing methods struggle to decouple these compounded dynamics from both representation and motion-modeling perspectives. On the representation side, aerial targets are typically tiny and heavily dominated by background texture; conventional feature extractors with static receptive fields and aggressive downsampling tend to submerge fine-grained boundaries and yield unstable embeddings under extreme scale changes and partial occlusion. On the motion/geometry side, most

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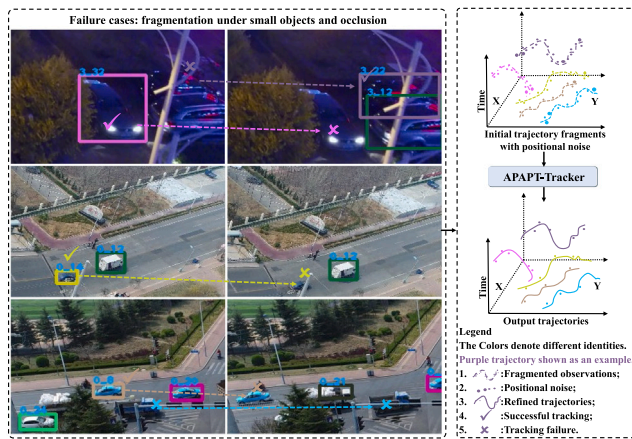


Fig. 1. Motivation of the proposed framework. The left side shows representative tracking failure cases caused by small-object difficulty and occlusion. The right side illustrates how fragmented trajectories with positional noise are refined into smoother and more complete output trajectories. Different initial trajectories are depicted with different amounts of local perturbation to reflect different noise severity. A unified legend is provided, where the purple trajectory is shown as an example and the same rules apply to the others.

tracking pipelines still rely on Kalman filtering with linear constant-velocity assumptions (Bewley et al., 2016; Aharon et al., 2022), which becomes brittle when the observed motion is the superposition of target dynamics and UAV ego-motion. While global trajectory prediction methods (e.g., Preformer MOT (Y. Wang et al., 2025)) can bridge gaps, they often require complex auxiliary training. World-coordinate modeling attempts such as WM-SORT (Y. Jiang et al., 2025) depend on strong priors (e.g., ground-plane constraints) and accurate camera parameter estimation, which are unreliable under signal instability and platform vibration (Liang et al., 2025). Moreover, high-frequency jitter in drone videos remains challenging: naive interpolation drifts on curvilinear motion induced by camera panning, whereas Gaussian-style smoothing introduces phase lag, causing boxes to detach from targets during rapid maneuvers (Chen et al., 2025; W. Liu et al., 2025).

Nevertheless, existing UAV trackers remain limited in jointly addressing both perception instability and trajectory inconsistency under coupled UAV/target dynamics. Specifically, (1) tiny targets under abrupt scale variation and background clutter still suffer from feature submergence and unstable appearance cues; and (2) motion coupling and high-frequency platform jitter often violate linear motion assumptions, where interpolation drifts on curvilinear motion while smoothing introduces temporal lag. As illustrated in Fig. 1, these issues commonly manifest as fragmented and noisy trajectory hypotheses, especially when targets are tiny and intermittently occluded. These observations motivate a unified solution that stabilizes both feature perception and trajectory geometry. To make this more explicit, the schematic illustration includes a unified legend and positional-noise annotations for all initial trajectories, where the amount of local perturbation reflects the relative severity of noise across different cases.

To address these limitations, we propose ADAPT-Tracker, which synergizes scale-aware representation learning and high-order geometric consistency recovery. First, we introduce Motion-Aware Receptive Field Modulation (MRFM). It explicitly separates fine-grained spatial details from coarse semantic context via a bi-granularity decomposition, and further employs a structure-preserving channel mixer to strengthen local topology without destructive pooling, thereby keeping targets discriminative under down-scaling and partial occlusion. Second, to restore consistent trajectories under ego-motion entanglement, we develop High-Order Geometric Stabilized Trajectory Refinement (H-GSTR) strategy. H-GSTR treats tracking refinement as a trajectory

manifold restoration problem: it reconstructs missing curvilinear trends through monotonicity-constrained temporal inpainting and suppresses observation jitter via local weighted least-squares regularization, effectively reducing noise without inducing temporal lag. In summary, our main contributions are as follows:

- We propose the ADAPT-Tracker (Adaptive Dynamic Perception and Trajectory-Consistent Network), a unified aerial MOT framework that couples scale-aware representation learning with geometry-driven trajectory stabilization to handle coupled UAV ego-motion and stochastic target motion. ADAPT-Tracker consistently outperforms the strongest competing method on UAVDT and VisDrone-MOT, improving by about +1.20/+2.00/+0.80 percentage points (HOTA/MOTA/IDF1) on UAVDT and +2.00/+3.30/+1.50 percentage points on VisDrone-MOT.
- We introduce Motion-Aware Receptive Field Modulation (MRFM) to alleviate the feature submergence of tiny targets under clutter. MRFM performs bi-granularity spatial decomposition to separate fine-grained local details from broader contextual semantics, and adopts a structure-preserving channel mixer to maintain spatial discriminability under down-scaling.
- We develop H-GSTR (High-Order Geometric Stabilized Trajectory Refinement) to repair fragmented tracklets caused by occlusions and missed detections under aggressive maneuvers. H-GSTR combines monotonicity-constrained bi-directional trajectory inpainting with locally weighted kinematic regularization to suppress non-linear observation jitter without introducing temporal lag. On UAVDT, it recovers 10.19% additional valid detections on average, peaking at 29.25% in high-viewpoint small-target scenes.

2. Related work

This section reviews the literature pertinent to our proposed network, categorized into three dimensions. First, we analyze the evolution of Multi-Object Tracking Paradigms (Section 2.1), emphasizing the transition from general-purpose methods to specialized architectures adapted for UAV constraints. Second, we examine Feature Representation Learning (Section 2.2), focusing on strategies designed to mitigate the semantic ambiguity of small-scale aerial targets. Finally, we discuss Trajectory Refinement Techniques (Section 2.3), evaluating motion modeling and post-processing approaches aimed at resolving trajectory fragmentation caused by occlusion and platform jitter.

2.1. Multi-object tracking paradigms and adaptability to UAVs

Multi-Object Tracking (MOT) has been dominated by the Tracking-by-Detection (TBD) paradigm (Fang et al., 2025), which decomposes video understanding into per-frame detection and temporal identity association. Early TBD pipelines, exemplified by SORT (Bewley et al., 2016) and DeepSORT (Wojke et al., 2017), combine Kalman filtering for motion prediction with appearance embeddings for ReID-based matching. (F. Liu et al., 2025) systematically reviewed remote sensing video tracking from datasets and metrics to model-based and deep-learning-based methods. Building upon this pipeline, subsequent studies have aimed to improve efficiency and robustness. For instance, JDE (Xu and Huang, 2024) and FairMOT (Zhang et al., 2021) unify detection and embedding learning to reduce inference latency, while MOSAIC-Tracker (Zou et al., 2025) mitigates occlusion-induced misses by exploiting low-confidence detections to enhance association continuity. Dc-MOT (Cheng et al., 2023) addresses motion blur caused by high-speed UAV/target motion and motion coupling between the UAV camera and target movement, and improves tracking by combining hybrid deblurring with motion compensation across adjacent frames. This line of work further highlights the importance of motion-aware tracking design under drone perspectives. (Ji et al., 2026) proposed a view-centric

homographic matching framework for moving-UAV MOT, showing that explicit geometric alignment across views can substantially improve association robustness in dynamic aerial scenes.

Beyond TBD, recent research has explored alternative paradigms to reduce reliance on hand-crafted association heuristics. MOTIP (Gao et al., 2025) proposes an end-to-end perspective by directly decoding identity-related outputs from detections. To better handle association ambiguity in complex scenes, SparseTrack (Z. Liu et al., 2025) introduces pseudo-depth cues and cascade matching under a sparse decomposition formulation. Meanwhile, Transformer-based trackers have gained increasing attention for global reasoning; for example, SparseDeTrack (Miao et al., 2026) adopts sparse tokens and removes the explicit ReID branch to balance computational cost and tracking accuracy.

However, directly transferring general-purpose paradigms to UAV viewpoints remains challenging. UAV videos exhibit severe scale variation, cluttered backgrounds, and frequent 6-DoF camera motion, which jointly induce non-linear observation noise and unstable appearances (Wu et al., 2025). In particular, tiny aerial targets (e.g., in VisDrone-MOT) often provide only weak and noisy discriminative cues, making both detector confidence and appearance embeddings less reliable for downstream association. Meanwhile, rapid ego-motion and platform jitter substantially deviate from the linear and smooth-motion assumptions underlying standard Kalman filtering, leading to inaccurate motion prediction and drifted gating regions.

To alleviate these UAV-specific issues, DMTrack (Fu et al., 2025) employs a Mamba-based deformable state space model to aggregate historical motion states under complex maneuvers, and MG-MOT (C.-Y. Yang et al., 2024) further leverages auxiliary metadata (e.g., GPS and altitude) to support long-term association in domain-specific settings. Despite these advances, existing UAV-oriented trackers often depend on stronger external priors (e.g., metadata) or implicitly assume that feature quality remains sufficiently stable, leaving two key challenges insufficiently addressed: (i) learning more reliable representations for tiny aerial targets at the feature extraction stage, and (ii) recovering trajectory continuity under occlusion and non-linear jitter at the association stage. These observations motivate our unified design that couples scale-aware feature learning with geometry-aware trajectory refinement.

2.2. Feature representation learning for small objects

In UAV and remote sensing videos, targets often occupy only a few pixels and are frequently embedded in cluttered backgrounds, leading to weak discriminative cues and unstable appearances (Q. Li et al., 2024, 2025; W. Zhang et al., 2025). Under this setting, the quality of feature representations strongly affects both detection reliability and identity association. Existing efforts on small-object feature enhancement (K. Liu et al., 2025; Li et al., 2026) can be broadly summarized into two lines: multi-scale and resolution enhancement, and feature interaction and fusion (Zhao et al., 2025).

This line aims to strengthen tiny-object representations by aggregating contextual cues across scales or recovering missing high-frequency details. For point-like targets in satellite videos, MCTracker (B. Wang et al., 2024) employs hybrid scale enhancement by cascading multi-level features to improve scale sensitivity. ArbiTrack (Chen et al., 2025a) introduces multi-scale context aggregation to handle arbitrary orientations under moving platforms. To alleviate physical resolution constraints, SuperMOT (Ren et al., 2025) integrates video super-resolution with pyramid deformable alignment, which can improve fine details but often increases computational cost and may depend on fixed scaling configurations, limiting generalization across diverse UAV scenes. Dist-Tracker (W. Wang et al., 2025) addresses scale-related instability from a metric learning perspective via a scale-aware Wasserstein distance and shape regularization. FWTrack (Fan et al., 2025)

leverages 3D wavelet transforms to construct hierarchical spatiotemporal representations, enriching frequency-domain cues with lightweight feature extraction. Despite demonstrated benefits in specific settings, many methods remain insufficiently content-adaptive when facing the extreme scale ranges and abrupt scale transitions commonly observed in UAV benchmarks such as VisDrone-MOT.

Another line focuses on coordinating complementary cues across tasks or modalities to reduce feature conflicts and improve association robustness. Interactive Attention Network (S. Ma et al., 2024) designs an interaction mechanism that explicitly models feature discrepancies between detection and tracking branches, facilitating information exchange. AMF-MOT (Cao et al., 2025) further proposes adaptive fusion with dynamic weight allocation to balance appearance and motion cues according to occlusion status, improving matching stability in challenging sequences. (Zhu et al., 2023) developed a spatio-temporal hierarchical feature transformer that fuses shallow spatio-temporal similarity cues with deeper semantic representations for UAV tracking. (Chen et al., 2025b) introduced a prompt-guided frequency-aware filtering and template-enhanced interaction framework for hyperspectral video tracking, emphasizing spectral-spatial-temporal coherence. From the perspective of difficult-target recognition, (Sun et al., 2026) proposed a consistency-reasoning strategy that exploits spatial-semantic priors to recover objects with limited visual evidence. These studies collectively highlight the importance of multi-level fusion, contextual reasoning, and occlusion-aware representation enhancement in remote sensing scenarios.

Despite recent advances, complex UAV scenes often amplify scale variation and background clutter, making tiny-object cues fragile. Existing enhancement modules may trade accuracy for additional computation, or provide limited content-adaptivity under abrupt scale changes. Therefore, we explore a motion-aware representation design that aims to retain local structures while adapting receptive fields, which is presented in our method section.

2.3. Trajectory refinement in aerial scenarios

Even with strong detection and association, aerial viewpoints frequently suffer from occlusions and ego-motion-induced non-linear jitter, which can lead to fragmented trajectories. Therefore, trajectory refinement and motion modeling are often used as complementary components to improve temporal continuity. Existing approaches mainly fall into two categories: motion model optimization and trajectory restoration.

A common restoration practice is linear interpolation, which can fill very short gaps but becomes unreliable under long occlusions or abrupt maneuvers, often producing over-simplified motion that violates realistic kinematic patterns. Beyond interpolation, many studies improve motion modeling by extending the Kalman Filter (KF). FastTrack (Liu et al., 2024) focuses on efficiency via parallel filtering and GPU acceleration for large-scale tracking. To better handle non-linear dynamics, ACNTrack (L. Zhang et al., 2025) introduces a neural Kalman filter to adapt noise parameters to motion states, but it may be data-dependent and sensitive to training distributions. SAGA-KF (Sani et al., 2024) leverages graph relations among targets for state estimation, while HP-Tracker (Z. Liu et al., 2025) applies higher-order numerical correction to reduce drift under missed detections. However, these KF variants can still struggle with the high-frequency, irregular jitter commonly observed in UAV videos, leading to delayed responses or oversmoothing.

For trajectory restoration, correlation-filter-based trackers such as improved KCF variants (Yin et al., 2025) incorporate confidence modeling (e.g., fuzzy logic) to stabilize prediction under partial observations. Learning-based recovery methods also emerge; for example, AIPT (Zhang et al., 2024) designs an information-guided module that

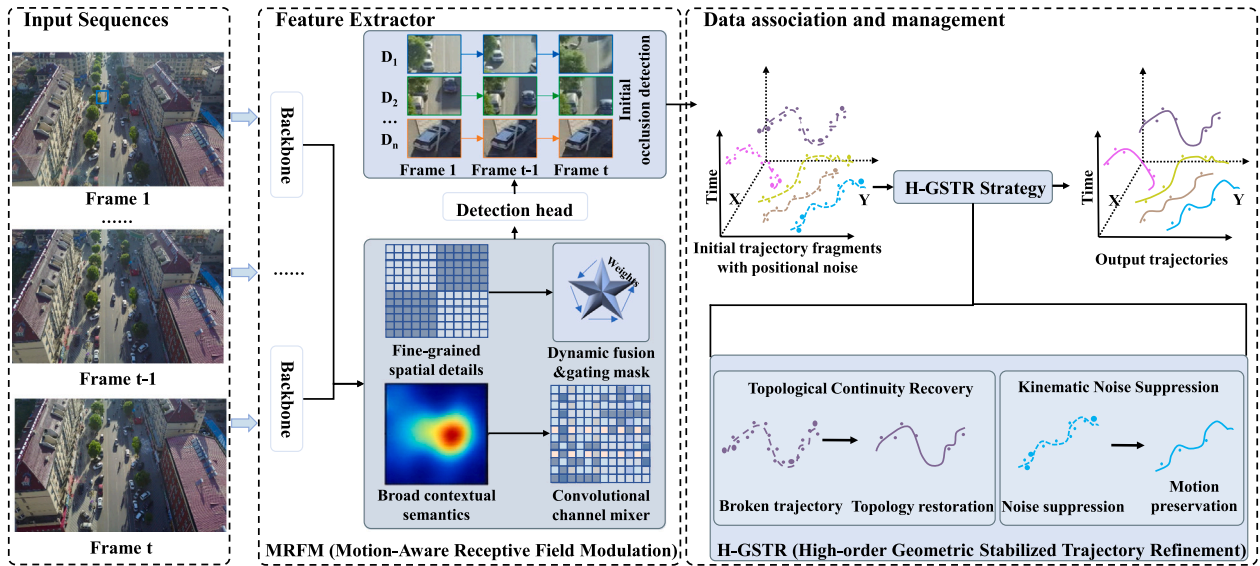


Fig. 2. The overall architecture of the proposed ADAPT-Tracker. The pipeline integrates two key components. (Middle) The Motion-Aware Receptive Field Modulation (MRFM) module generates scale-aware representations through bi-granularity decomposition, fusion within the gating operator, and multiplicative context modulation, followed by post-modulation channel mixing. (Right) The High-order Geometric Stabilized Trajectory Refinement (H-GSTR) module serves as a post-processing component to recover bounded topological breaks and suppress kinematic noise, thereby improving trajectory continuity and geometric stability.

recovers missing tracklets by exploiting motion priors learned from historical states. (Xiao et al., 2025) proposed a sparse satellite video tracking framework with hierarchical spatio-temporal fusion and Gaussian-based trajectory association to reduce observational drift and accumulated error. (Wang et al., 2026) introduced a joint learning framework of historical trajectory guidance and identity prediction, demonstrating that historical motion flow can enhance long-term continuity under challenging vehicle scenarios. These studies further confirm that trajectory continuity in remote sensing videos benefits from geometry-aware restoration and historical-motion-guided refinement rather than from simple interpolation alone. In addition, tracking studies based on motion information and perturbation-aware trajectory reconstruction enrich the discussion of motion-aware tracking in dynamic environments. For instance, observer-based dual hybrid nonfragile tracking control (C. Liu et al., 2025) for satellite swarm reconstruction emphasizes robust trajectory tracking under coexisting disturbances and provides a meaningful perspective on relative motion-aware tracking and trajectory reconstruction. Despite progress, existing refinement schemes often depend on specific parametric assumptions or require careful tuning to balance smoothness and responsiveness in aerial scenarios. These limitations motivate a geometry-aware refinement strategy that aims to recover trajectory continuity under occlusion while stabilizing non-linear jitter. In this work, we introduce *High-order Geometric Stabilized Trajectory Refinement (H-GSTR)* to address these challenges, with details provided in the method section.

3. Methodology

3.1. Overview of the proposed method

For clarity and mathematical consistency, the principal symbols, operators, and variables used in the methodology are summarized in Table 1. Unless otherwise specified, each symbol retains the same meaning throughout the paper.

As illustrated in Fig. 2, we propose a unified tracking-by-detection framework tailored to the complex spatiotemporal dynamics in UAV surveillance. Our method is built upon the BoT-SORT framework. In our implementation, the camera motion compensation (CMC) module is retained. The ReID branch is also supported and can be enabled explicitly through the runtime `--reid` option. In the main

reported testing configuration on both UAVDT and VisDrone, inference is conducted with ReID enabled, while the dedicated ablation study further toggles this option to quantify its practical contribution. Therefore, unless otherwise specified, each experiment follows the exact BoT-SORT configuration indicated in the corresponding table. To alleviate the scale-resolution challenge, we insert *Motion-Aware Receptive Field Modulation (MRFM)* before the detection head to enhance scale-robust feature representations. MRFM decomposes features into complementary local-detail and global-context components and adaptively modulates their contributions to improve multi-scale perception. After generating preliminary tracklets with a BoT-SORT-based association strategy, we apply *High-order Geometric Stabilized Trajectory Refinement (H-GSTR)* as a post-processing module. H-GSTR aims to repair short bounded interruptions within established trajectories and stabilize ego-motion-induced non-linear jitter via geometric refinement and kinematic regularization, producing trajectories that are more geometrically consistent. Importantly, H-GSTR operates directly on these tracker-generated trajectories after data association, rather than on benchmark ground-truth trajectories. H-GSTR does not introduce an additional cross-ID re-association stage; instead, it refines each associated trajectory independently through bounded gap restoration and local stabilization. At each frame t , the detector outputs $D_t = \{(O_t^{(i)}, s_t^{(i)})\}_{i=1}^{N_t}$, where $O_t^{(i)} \in \mathbb{R}^4$ denotes the box state. We adopt a BoT-SORT-style association backbone to form preliminary tracklets, which are further refined by H-GSTR. The inference pipeline of ADAPT-Tracker is summarized in Algorithm 1.

3.2. Motion-aware receptive field modulation

Fig. 3 illustrates the architecture of *Motion-Aware Receptive Field Modulation (MRFM)*. MRFM is designed to mitigate the scale-resolution challenge in UAV imagery by encoding features with adaptive receptive fields. Rather than serving as a generic multi-kernel feature extractor, MRFM is motivated by the feature submergence problem in aerial tracking, where tiny targets are easily overwhelmed by cluttered backgrounds and unstable scale variation. The module adopts a dual-branch design: a local branch uses 3×3 depth-wise convolutions (DWConv) to preserve fine-grained structures that are critical for tiny targets, while a global branch employs 7×7 DWConv to capture broader contextual semantics under a larger receptive field. A context-gated modulation

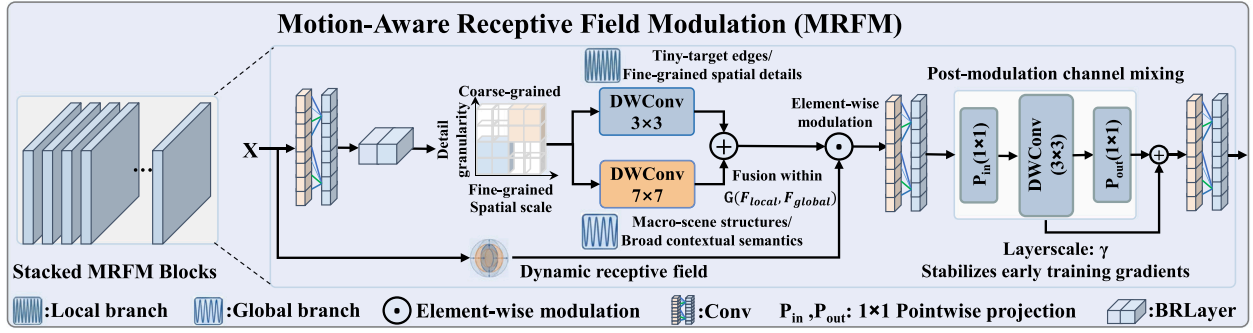


Fig. 3. Schematic of the Motion-Aware Receptive Field Modulation (MRFM). The projected input feature is processed by two parallel depth-wise convolution branches to extract fine-grained local details and broader contextual semantics, respectively. These branch responses are first fused within the gating operator to form a context-aware mask, which is then applied to the projected feature through element-wise multiplication, consistent with Eq. (4). The post-modulation channel mixer is subsequently applied to the modulated feature as a structure-preserving transformation.

Table 1

Main mathematical symbols and operators used in the methodology.

Symbol	Description
General Formulation	
$I(t)$	Observed image sequence at time t
$S(t)$	Latent clean scene
$\mathcal{T}_{ego}(\cdot)$	Ego-motion transformation operator
$D_t = \{(O_t^{(i)}, s_t^{(i)})\}_{i=1}^{N_t}$	Detections at frame t , where denotes the detection confidence
$O_t, s_t \in \mathbb{R}^4$	Observed box state and latent clean trajectory state, respectively
m_t	Binary observation mask indicating whether a target is observed
MRFM	
(Section 3.2) X, Y, Y^*	Input feature map, latent modulated feature, and optimal conceptual solution
λ_1, λ_2	Weights balancing local-detail and global-context regularization
∇_l, ∇_g	Local- and global-scale differential operators
$\phi(\cdot)$	Input projection function mapping features into an intermediate embedding space
$\psi(\cdot)$	Output projection function after gated modulation
F_{local}, F_{global}	Local-detail and global-context branch features
$DWC_{k \times k}$	Depth-wise convolution with kernel size $k \times k$
$\mathcal{G}(\cdot)$	Context-gated fusion operator
$\odot$	Hadamard product (element-wise multiplication)
P_{in}, P_{out}	Pointwise convolutions for local channel interaction
σ, γ	Nonlinear activation and LayerScale factor
H-GSTR	
(Section 3.3) $\hat{T}$	Refined trajectory after geometric stabilization
$H(t)$	Piecewise cubic Hermite interpolant for gap recovery
$h_{ij}(s)$	Cubic Hermite basis functions with normalized time $s \in [0, 1]$
d_k, d_{k+1}	Endpoint tangents used in Hermite interpolation
M	Temporal window radius in KNS
$y \in \mathbb{R}^{2M+1}$	Local observation vector within the sliding window
V, W	Vandermonde matrix, diagonal weight matrix
$\hat{c}$	Estimated polynomial coefficients
$\hat{O}_t$	Refined state at time t after KNS
w_{SG}, e_1	Centered Savitzky–Golay kernel and first canonical basis vector

Algorithm 1 Inference Pipeline of ADAPT-Tracker

Require: Video sequence $\mathcal{V} = \{I_t\}_{t=1}^T$; detector with MRFM

Ensure: Refined trajectories $\hat{\mathbb{T}}$

- 1: **Initialize** track set $\mathbb{T}_0 \leftarrow \emptyset$
- 2: **for** $t = 1$ **to** T **do**
- 3: **Stage 1: Scale-Aware Detection with MRFM** (Sec. 3.2)
- 4: Run detector with MRFM-enhanced features to obtain $D_t = \{(O_t^{(i)}, s_t^{(i)})\}_{i=1}^{N_t}$
- 5: **Stage 1: Association Backbone (Kalman + Hungarian)**
- 6: Predict track states via Kalman filter: $\hat{O}_t^j \leftarrow \text{KF-Predict}(\mathcal{T}^j)$ for each $\mathcal{T}^j \in \mathbb{T}_{t-1}$
- 7: Construct cost/similarity matrix between $\{\hat{O}_t^j\}$ and D_t (optionally using τ for confidence splitting)
- 8: Update tracklets by Hungarian matching and KF update to obtain $\mathbb{T}_t$
- 9: **end for**
- 10: **Stage 2: High-Order Geometric Refinement (H-GSTR)** (Sec. 3.3)
- 11: $\hat{\mathbb{T}} \leftarrow \mathbb{T}_T$
- 12: **for each** trajectory $\mathcal{T}^j \in \hat{\mathbb{T}}$ **do**
- 13: **TCR: Topological Continuity Recovery** (Sec. 3.3.2)
- 14: **if** $\mathcal{T}^j$ contains short bounded missing intervals **then**
- 15: Recover missing states within valid intra-track gaps using monotonicity-constrained Hermite interpolation. (Eqs. (7)–(9))
- 16: **end if**
- 17: **KNS: Kinematic Noise Suppression** (Sec. 3.3.3)
- 18: Construct Vandermonde matrix V (Eq. (10)) and compute WLS/SG kernel
- 19: Refine states by centered convolution (Eq. (12))
- 20: **end for**
- 21: **return** $\hat{\mathbb{T}}$

unit then fuses the two branches to generate an input-adaptive context mask, which modulates the projected feature to balance local details and global awareness.

3.2.1. Problem formulation: The scale–resolution challenge

In UAV tracking, targets often occupy very limited pixels (e.g., $< 16 \times 16$) and appear under cluttered backgrounds with rapid viewpoint changes. We model the observed image sequence $I(t)$ as a degraded observation of the latent scene $S(t)$:

$$I(t) = \mathcal{T}_{ego}(t)(S(t)) + \mathcal{N}_{sensor}(t) + \epsilon_{motion}(t), \quad (1)$$

where $\mathcal{T}_{ego}(t)$ denotes the ego-motion transformation, $\mathcal{N}_{sensor}(t)$ represents sensor noise, and $\epsilon_{motion}(t)$ summarizes motion-induced degradations. Fixed receptive fields in standard CNNs may be insufficient to accommodate such scale-dependent degradations, motivating an adaptive modulation mechanism.

3.2.2. Bi-granularity feature decomposition

Drawing inspiration from scale-space modeling, we first introduce a conceptual objective that characterizes the desired trade-off among feature fidelity, local-detail preservation, and global-context smoothing at different granularities:

$$\mathbf{Y}^* = \arg \min_{\mathbf{Y}} \{ \|\mathbf{X} - \mathbf{Y}\|_2^2 + \lambda_1 \|\nabla_l \mathbf{Y}\|_2^2 + \lambda_2 \|\nabla_g \mathbf{Y}\|_2^2 \}, \quad (2)$$

Here, $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$ denotes the input feature map, $\mathbf{Y}$ denotes the latent scale-modulated representation to be estimated, and $\lambda_1, \lambda_2 \geq 0$ are balancing coefficients that control the relative strengths of the local-detail and global-context regularization terms, respectively. The operators ∇_l and ∇_g denote differential operators associated with local and global receptive fields. Eq. (2) is now explicitly described as a conceptual, motivation-oriented variational objective, which characterizes the desired balance among fidelity, local-detail preservation, and global-context smoothing. We then state clearly that Eqs. (3)–(4) are a lightweight neural instantiation of this principle, rather than a strict analytical derivation. Since directly solving Eq. (2) inside a deep detector is impractical, we instantiate it with a lightweight dual-branch neural approximation as follows:

$$\begin{aligned} \mathbf{F}_{local} &= DWC_{3 \times 3}(\phi(\mathbf{X})), \\ \mathbf{F}_{global} &= DWC_{7 \times 7}(\phi(\mathbf{X})), \end{aligned} \quad (3)$$

In Eq. (3), $\phi(\cdot) : \mathbb{R}^{C \times H \times W} \rightarrow \mathbb{R}^{C' \times H \times W}$ denotes a learnable linear projection function that maps the input feature map into an intermediate embedding space before branch-specific processing. The local branch output $\mathbf{F}_{local}$ emphasizes fine-grained local structures, whereas the global branch output $\mathbf{F}_{global}$ captures broader contextual dependencies under a larger receptive field. We then perform context-gated modulation, in which the two branch responses are first fused within the gating operator to form a context-aware mask, and the resulting mask is subsequently used to modulate the projected feature.

$$\mathbf{Y}_{mod} = \psi(\phi(\mathbf{X}) \odot \mathcal{G}(\mathbf{F}_{local}, \mathbf{F}_{global})), \quad (4)$$

In Eq. (4), $\odot$ denotes the Hadamard product, $\mathcal{G}(\mathbf{F}_{local}, \mathbf{F}_{global})$ denotes a context-gated fusion operator that aggregates the complementary local and global responses into a unified context-aware mask, and $\psi(\cdot)$ denotes the output linear projection used to generate the modulated feature. We further emphasize that Eq. (2) is a motivation-oriented formulation rather than a strict analytical derivation of Eqs. (3)–(4): the additive regularization terms in Eq. (2) conceptually describe complementary local/global priors, whereas Eq. (4) gives the actual feed-forward realization through gated element-wise modulation. In the present MRFM implementation, the operator $\mathcal{G}(\mathbf{F}_{local}, \mathbf{F}_{global})$ is instantiated by first fusing the local-detail and global-context branches to form a unified context-aware mask, which is then applied to the projected feature $\phi(\mathbf{X})$ via the Hadamard product in Eq. (4). The post-modulation channel mixer is subsequently applied to the modulated feature as a structure-preserving transformation, rather than operating as a parallel processor of the two branches.

3.2.3. Structure-preserving channel evolution

Channel mixing modules such as Squeeze-and-Excitation (SE) typically rely on Global Average Pooling (GAP) to summarize spatial responses. While effective for many recognition tasks, GAP removes spatial variation by global aggregation, which may weaken localized cues for tiny targets in UAV tracking. Motivated by this observation, we replace GAP with a local interaction mechanism to preserve fine-grained structures during channel evolution:

$$\mathbf{Y}_{out} = \mathcal{P}_{out}(\sigma(DWC_{3 \times 3}(\sigma(\mathcal{P}_{in}(\mathbf{Y}_{mod}))))), \quad (5)$$

where $\mathcal{P}_{in/out}$ are pointwise convolutions and σ is a non-linear activation. From a spatial-detail preservation perspective, local depth-wise convolution retains spatially varying responses and can better preserve fine-grained structures associated with tiny-object boundaries and local shape cues. To stabilize optimization, we adopt LayerScale with a learnable scalar factor γ , initialized to 10^{-6} , and obtain the final MRFM output as $\mathbf{Y}_{final} = \mathbf{X} + \gamma \cdot \mathbf{Y}_{out}$. Compared with conventional multi-scale convolution modules, the key innovation of MRFM lies not merely in using different kernel sizes, but in combining bi-granularity branch decomposition with context-gated modulation for input-adaptive receptive-field adjustment. Moreover, unlike GAP-based channel enhancement mechanisms that tend to collapse spatial variation through global squeezing, the proposed structure-preserving channel mixer enhances channel interaction through local depth-wise processing while retaining the fine-grained spatial topology that is important for tiny-object perception in UAV scenes.

3.3. High-order geometric stabilized trajectory refinement

Fig. 4 illustrates the intuition behind H-GSTR (High-Order Geometric Stabilized Trajectory Refinement). The novelty of H-GSTR lies in a problem-driven trajectory restoration formulation for aerial MOT, where two coupled degradation modes coexist: short bounded trajectory interruptions and high-frequency non-linear observation jitter. To address these two failure modes in a coordinated manner, H-GSTR explicitly decomposes refinement into Topological Continuity Recovery (TCR) for bounded gap restoration and Kinematic Noise Suppression (KNS) for zero-phase local stabilization. This structured coupling makes H-GSTR more suitable for aerial trajectory refinement than generic interpolation-only or smoothing-only post-processing pipelines.

H-GSTR is designed as a trajectory-level refinement module that operates on the association results produced by the upstream tracker. In practice, it is applied after frame-by-frame association and uses the established track hypotheses returned by the upstream tracker as its refinement input. Its role is to improve the resulting trajectories by repairing recoverable short interruptions and suppressing local geometric jitter, thereby enhancing temporal continuity and trajectory completeness. Under this design, long-term identity re-association remains the responsibility of the upstream association stage, while H-GSTR focuses on improving the completeness and consistency of the associated trajectories. In the input, dashed polylines indicate fragmented track segments (e.g., under small objects and occlusion), while solid curves denote intact portions; dense scattered dots represent high-frequency observation noise from UAV jitter and detection perturbations. For clarity, we visualize two components separately: TCR (Topological Continuity Recovery) and KNS (Kinematic Noise Suppression). The green trajectory exemplifies how TCR reconnects fragmented observations by taking the two valid observations that bound a short missing interval within the same associated trajectory as interpolation endpoints, and then performing geometric topological inpainting between them. In contrast, the blue trajectory highlights how KNS suppresses jitter within a sliding temporal window while preserving the motion trend.

3.3.1. Problem formulation

Given a trajectory $\mathcal{T} = \{O_1, O_2, \dots, O_T\}$ produced by a baseline tracker in UAV videos, each observation $O_t \in \mathbb{R}^4$ denotes the bounding box state at time t (e.g., center, width, height). Due to occlusions and platform-induced jitter, $\mathcal{T}$ can be incomplete and noisy. We model the observation as

$$O_t = m_t(\mathbf{s}_t + \epsilon_t), \quad m_t \in \{0, 1\}, \quad (6)$$

where $\mathbf{s}_t \in \mathbb{R}^4$ is the latent clean state, m_t indicates whether the target is observed at time t , and ϵ_t denotes localization noise.

In practice, trajectory noise in aerial tracking mainly arises from two coupled sources. First, trajectory incompleteness is mainly caused by

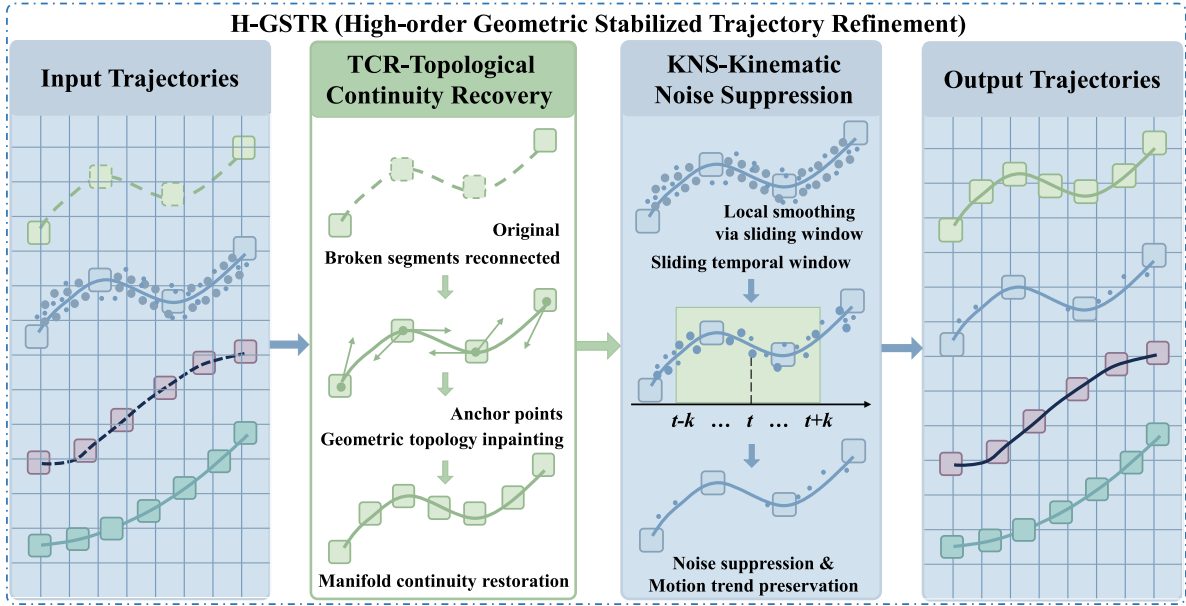


Fig. 4. Mechanism of H-GSTR. Dashed lines denote fragmented segments and solid curves denote intact portions; scattered dots indicate high-frequency observation noise. TCR restores short bounded missing intervals within the same associated trajectory by using the two observed endpoints of the gap for geometric topological inpainting (green example), while KNS suppresses jitter in a sliding temporal window and preserves motion trends (blue example). The two components operate together to refine the output trajectories.

occlusion, temporary missed detections, image degradation, and tracking instability, which leads to missing observations and broken temporal continuity. Second, localization noise is mainly caused by UAV ego-motion, platform vibration, abrupt viewpoint changes, and detector perturbations, which introduces high-frequency geometric fluctuation even when detections remain available. For simplicity, we approximate the jitter term as zero-mean noise (e.g., $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$), which primarily affects short-term geometry. Here, s_t is introduced only as a conceptual latent clean state in the degradation model. It is not a benchmark ground-truth trajectory, nor is it used as a direct fitting target in H-GSTR. These two degradation sources affect the trajectory in different ways: missing observations mainly lead to fragmented topology and discontinuous temporal support, whereas localization noise mainly causes unstable box centers, fluctuating widths/heights, and distorted short-term motion trends. As a result, the observed trajectory may become both incomplete and geometrically inconsistent.

Our objective is to reconstruct a refined trajectory $\hat{\tau}$ by repairing short interruptions within an established track and suppressing high-frequency jitter, thereby improving temporal continuity and geometric consistency. In this formulation, occlusion is highlighted as a representative source of trajectory interruption in aerial tracking, but it is not the only one: recoverable gaps may also arise from temporary missed detections, image degradation, or tracking instability. Importantly, H-GSTR is designed as a bounded post-processing refinement module, where recovery is performed only within the observed temporal span of a trajectory and only when the missing interval is enclosed by two valid observations of the same track. Under this formulation, Topological Continuity Recovery (TCR) is responsible for bounded gap filling, while Kinematic Noise Suppression (KNS) is responsible for local stabilization. Their roles are complementary rather than redundant: TCR restores fragmented trajectory topology, whereas KNS suppresses short-term kinematic perturbation without introducing temporal lag.

In practice, bounded gap recovery is triggered when two valid observations of the same associated trajectory are separated by a short missing interval that does not exceed the predefined temporal threshold. And benchmark ground-truth annotations are used only for evaluation under standard MOT protocols, such as HOTA, MOTA, and IDF1, rather than for guiding the H-GSTR refinement itself. Therefore,

any annotation noise in the benchmark affects evaluation uniformly across methods, but does not serve as a reconstruction target for H-GSTR.

3.3.2. Topological Continuity Recovery (TCR)

TCR estimates missing states $\hat{O}_t$ only for short intra-track gaps bounded by two valid observations within the same trajectory. Linear interpolation is lightweight but may oversimplify motion under longer interruptions, while standard cubic splines can introduce oscillations (Runge's phenomenon). We therefore construct TCR as a monotonicity-preserving Hermite interpolator for bounded gap recovery, where interpolation is applied only when the temporal gap remains within a predefined maximum length. This restriction reduces the risk of over-aggressive reconstruction for interruptions that are not reliably recoverable. The rationale for adopting PCHIP in TCR is not to increase interpolation order, but to obtain a shape-preserving bounded recovery scheme for short trajectory gaps. In aerial MOT, tiny-object trajectories are highly sensitive to interpolation overshoot, since even modest geometric distortion may significantly affect box quality and subsequent association. Compared with higher-order polynomial interpolation, PCHIP provides a more suitable inductive bias by preserving local shape and monotonicity, thereby reducing oscillatory artifacts during bounded gap recovery.

Let $s = \frac{t-t_k}{h_k} \in [0, 1]$ be the normalized time, where $h_k = t_{k+1} - t_k$. The interpolant $H(t)$ is expressed in the Hermite basis:

$$H(t) = h_{00}(s)O_k + h_{10}(s)h_k d_k + h_{01}(s)O_{k+1} + h_{11}(s)h_k d_{k+1}, \quad (7)$$

where O_k, O_{k+1} are observed endpoints and d_k, d_{k+1} are first-order derivatives (tangents). The cubic Hermite basis functions $h_{ij}(s)$ are

$$\begin{aligned} h_{00}(s) &= 2s^3 - 3s^2 + 1, \\ h_{10}(s) &= s^3 - 2s^2 + s, \\ h_{01}(s) &= -2s^3 + 3s^2, \\ h_{11}(s) &= s^3 - s^2. \end{aligned} \quad (8)$$

To promote physically plausible trends, we enforce a monotonicity constraint on the tangents following the Fritsch–Carlson criterion. Let

the secant slope be $\delta_k = (O_{k+1} - O_k)/h_k$. For monotone segments, the tangent d_k is computed by a weighted harmonic mean:

$$\frac{1}{d_k} = \frac{1}{w_1 + w_2} \left(\frac{w_1}{\delta_{k-1}} + \frac{w_2}{\delta_k} \right), \quad (9)$$

with $w_1 = 2h_k + h_{k-1}$ and $w_2 = h_k + 2h_{k-1}$. If $\text{sgn}(\delta_{k-1}) \neq \text{sgn}(\delta_k)$, we set $d_k = 0$. TCR reconnects fragmented observations through endpoint-bounded interpolation within an associated trajectory. Specifically, when two valid observations of the same track are separated by a short missing interval, these two observations are treated as the endpoints of a recoverable gap, and the missing states between them are reconstructed by PCHIP-based interpolation. In the manuscript, occlusion is highlighted as a representative cause of such interruptions, but the practical scope is broader: recoverable short gaps may also arise from temporary missed detections, image degradation, or tracking instability. Therefore, TCR should be understood as a conservative restoration module for short bounded intra-track interruptions, rather than an explicit occlusion detector.

More specifically, a missing interval is considered recoverable only when two observations belong to the same associated track, the gap lies within the observed temporal support of that trajectory, and the gap length does not exceed a predefined maximum threshold. This threshold is treated as a dataset-level hyperparameter rather than a universal constant, and is fixed for all sequences within each benchmark. In our experiments, we use `max_gap=32` for UAVDT and `max_gap=25` for VisDrone. The slightly larger threshold on UAVDT reflects its relatively more stable short-gap continuity, whereas the more conservative threshold on VisDrone helps reduce the risk of erroneous recovery under denser clutter, more frequent occlusion, and stronger small-target interference.

In implementation, trajectory refinement is applied independently to each associated track ID after data association. Missing frames are reindexed only within the temporal support of that trajectory, and interpolation is performed under an explicit maximum-gap constraint. Any interruption exceeding the allowed length remains unreconstructed, and no extrapolation is performed before the first or after the last valid observation. This conservative design ensures that H-GSTR mainly repairs short, recoverable intra-track interruptions rather than generating arbitrary missing objects.

3.3.3. Kinematic Noise Suppression (KNS)

After reconnecting fragmented trajectories, we further suppress the observation jitter ϵ_t . In aerial videos, causal smoothing (e.g., one-sided Gaussian averaging) can introduce temporal lag, causing the refined boxes to trail the target under abrupt maneuvers. We therefore formulate KNS as a local weighted least-squares (WLS) projection, which yields a centered (zero-phase) refinement for post-processing.

We model the observed sequence as the sum of a latent kinematic signal and noise, i.e., $O_t = S_t + \epsilon_t$, where ϵ_t is approximated as zero-mean perturbation. For a center time t and window radius M , we collect a local observation vector $\mathbf{y} \in \mathbb{R}^{2M+1}$ with entries $y_i = O_{t+i}$, $i \in [-M, M]$. Assuming that the local trajectory within the temporal window can be approximated by an n th order polynomial $p(\tau) = \sum_{j=0}^n c_j \tau^j$, where τ denotes the local temporal coordinate centered at the current frame and n controls the approximation order, we construct the Vandermonde matrix $\mathbf{V} \in \mathbb{R}^{(2M+1) \times (n+1)}$ as:

$$\mathbf{V} = \begin{bmatrix} (-M)^0 & (-M)^1 & \dots & (-M)^n \\ \vdots & \vdots & \ddots & \vdots \\ 0^0 & 0^1 & \dots & 0^n \\ \vdots & \vdots & \ddots & \vdots \\ M^0 & M^1 & \dots & M^n \end{bmatrix}. \quad (10)$$

We estimate the coefficients via a local WLS objective

$$\hat{\mathbf{c}} = \arg \min_{\mathbf{c}} \|\mathbf{W}(\mathbf{y} - \mathbf{V}\mathbf{c})\|_2^2, \quad (11)$$

where $\mathbf{W} = \text{diag}(w_{-M}, \dots, w_M)$ assigns higher weights to samples closer to the center (e.g., $w_i = \exp(-\frac{i^2}{2\sigma_w^2})$). This convex problem admits the

closed-form solution $\hat{\mathbf{c}} = (\mathbf{V}^T \mathbf{W} \mathbf{V})^{-1} \mathbf{V}^T \mathbf{W} \mathbf{y}$. The refined state at time t corresponds to the polynomial value at $\tau = 0$, i.e., $\tilde{O}_t = p(0) = c_0$. Let $\mathbf{H} = (\mathbf{V}^T \mathbf{W} \mathbf{V})^{-1} \mathbf{V}^T \mathbf{W}^2$; then the first row of $\mathbf{H}$ defines a convolution kernel $\mathbf{w}_{SG}$, yielding

$$\tilde{O}_t = \mathbf{e}_1^T \hat{\mathbf{c}} = \sum_{k=-M}^M \mathbf{w}_{SG}(k) \cdot O_{t+k}, \quad (12)$$

where $\mathbf{e}_1 = [1, 0, \dots, 0]^T$. Since the kernel is centered, the refinement is zero-phase for post-processing. Moreover, the projection exactly preserves polynomial trends up to degree n within the window, while attenuating high-frequency jitter. In practice, since $O_t \in \mathbb{R}^4$, we apply the above WLS/SG refinement independently to each component of the 4D box state (i.e., we construct $\mathbf{y}$ per dimension and perform element-wise filtering), yielding $\tilde{O}_t \in \mathbb{R}^4$.

4. Experiments

In this section, to evaluate the performance of the proposed method in aerial scenarios, we utilize two widely recognized UAV-based benchmarks: VisDrone-MOT and UAVDT. These datasets are characterized by distinct challenges inherent to drone surveillance, including rapid camera motion, small object scales, and high-density occlusion.

4.1. Datasets

VisDrone Dataset: VisDrone is a large-scale UAV benchmark family that supports multiple tasks such as object detection and multi-object tracking. In this paper, we use the multi-object tracking benchmark (VisDrone-MOT) and, for simplicity, refer to it as *VisDrone* unless otherwise stated. Under the commonly used VisDrone-MOT evaluation protocol (e.g., the MOT2019 setting), the benchmark contains 79 video clips with 33,366 frames, split into 56 training clips, 7 validation clips, and 16 testing/evaluation clips. The MOT evaluation typically focuses on five categories: pedestrian, car, van, bus, and truck. This benchmark is challenging due to frequent small objects, dense scenes, and rapid viewpoint changes in drone videos.

UAVDT Dataset: UAVDT is a UAV benchmark for detection and tracking in complex urban traffic scenes. It contains 100 video sequences (about 80,000 representative frames) captured from UAV platforms, and provides scene attributes including flying altitude, camera view, and weather condition. UAVDT is particularly challenging due to heavy traffic density, long-term occlusions, and large resolution changes caused by altitude and viewpoint variations.

4.2. Evaluation metrics

We adopt standard evaluation protocols for multi-object tracking, utilizing HOTA, MOTA, and IDF1 as the primary performance indicators. These are complemented by detailed diagnostics, including False Positives (FP), False Negatives (FN), Identity Switches (IDs), Mostly Tracked (MT), and Mostly Lost (ML) trajectories.

HOTA (Higher Order Tracking Accuracy): As a unified metric balancing detection and association, HOTA is defined as the geometric mean of the detection score (DetA) and association score (AssA):

$$\text{HOTA} = \sqrt{\text{DetA} \cdot \text{AssA}} \quad (13)$$

MOTA (Multiple Object Tracking Accuracy): MOTA primarily measures detection performance by penalizing False Positives (FP), False Negatives (FN), and Identity Switches (IDs) over all frames t :

$$\text{MOTA} = 1 - \frac{\sum_t (\text{FN}_t + \text{FP}_t + \text{IDs}_t)}{\sum_t \text{GT}_t} \quad (14)$$

where GT_t denotes the number of ground truth objects.

IDF1 (Identification F1 Score): IDF1 evaluates identity consistency based on the F1 score of correct ID matches (IDTP) versus false ID matches (IDFP, IDFN):

$$\text{IDF1} = \frac{2 \cdot \text{IDTP}}{2 \cdot \text{IDTP} + \text{IDFP} + \text{IDFN}} \quad (15)$$

4.3. Implementation details

Experiments are implemented in PyTorch 2.5.0 on the NVIDIA RTX A6000 GPU. The detector in our framework is built on the Ultralytics YOLOv8s architecture. The baseline detector uses the standard YOLOv8s model, while the improved detector adopts our MRFM-enhanced architecture within the same framework. To ensure a fair comparison, both the baseline detector and the MRFM-enhanced detector are initialized from the same pretrained weights.

Unless otherwise specified, the detector-side training setup is kept consistent across the compared models. On both VisDrone and UAVDT, the detector is trained for 30 epochs with a batch size of 8 and an input resolution of 1280×1280 . Optimization is performed via SGD (momentum: 0.937, weight decay: 0.0005) with an initial learning rate of 0.001 under a cosine annealing schedule. Thus, the detector-side comparison is conducted under a matched architecture family, matched pretrained initialization, and matched main training configuration, with the principal difference being whether the proposed MRFM module is introduced.

At the tracking stage, both datasets use the same BoT-SORT-based tracking pipeline, and the main reported testing configuration enables the ReID branch through the runtime `--reid` option. The final tracking performance is evaluated using the same TrackEval-based workflow. Secondary implementation details follow the corresponding default framework configuration unless otherwise specified.

4.4. Comparison with advanced methods

To comprehensively verify the robustness of our method, we conduct extensive evaluations on the VisDrone-MOT and UAVDT benchmarks against extensive state-of-the-art algorithms, categorized into four distinct groups. First, we evaluate novel representation paradigms (2025) that introduce cutting-edge innovations, such as the segmentation-based SAM2MOT (J. Jiang et al., 2025), field-aware DFA-MOT (Zheng et al., 2025), and the track-centric TrackTrack (Shim et al., 2025). Second, we benchmark against spatio-temporal optimizations (2024–2025), including the motion-guided MGTrack (Ren et al., 2024) for high-speed targets, AdaptTrack (K. Li et al., 2024) utilizing adaptive correlation, and the real-time framework TStm (Zhang and Ye, 2025). Third, we examine UAV-specific architectures tailored for aerial challenges, comprising UAVMOT (Liu et al., 2022) and DroneMOT (P. Wang et al., 2024) for view variations, alongside UCMCTrack (Yi et al., 2024), STCMOT (J. Ma et al., 2024) for camera compensation and spatio-temporal cohesion and occlusion-robust MOSAIC-Tracker (Zou et al., 2025). Finally, we include classical and general Baselines, ranging from foundational trackers like SORT (Bewley et al., 2016) and DeepSORT (Wojke et al., 2017) to widely adopted general-purpose SOTAs such as FairMOT (Zhang et al., 2021) and ByteTrack (Zhang et al., 2022), which optimize association, as well as BoT-SORT (Aharon et al., 2022) and OCSORT (Cao et al., 2023) for robust camera motion modeling.

4.4.1. Comparison on the UAVDT dataset

The comparative results on the UAVDT benchmark are reported in Table 2. We primarily evaluate performance using the three standard MOT metrics (HOTA, MOTA, and IDF1). The remaining statistics (FP, FN, IDs, MT, and ML) are reported as reference diagnostics to interpret these main metrics. Our proposed method achieves a new state-of-the-art (SOTA) performance across all primary metrics, demonstrating exceptional robustness in complex aerial scenarios. Specifically, it attains 65.766% HOTA, 72.314% MOTA, and 82.499% IDF1. As illustrated in the results, traditional generic trackers, such as ByteTrack (Zhang et al., 2022) and BoT-SORT (Aharon et al., 2022), struggle to maintain consistent performance due to the domain gap, lagging behind our method. Furthermore, compared to recent domain-specific algorithms published in 2024 and 2025, our method maintains a distinct advantage. For instance, while SAM2MOT (J. Jiang et al., 2025) and MOSAIC-Tracker

(Zou et al., 2025) exhibit strong competitiveness in specific metrics, our approach surpasses the strongest competing method by about 1.20% in HOTA and 2.00% in MOTA and 0.80% in IDF1. This indicates that our method provides a more balanced solution for the conflicting requirements of detection precision and association stability.

The quantitative gains directly reflect the effectiveness of our method. The substantial improvement in MOTA over the second best) and the reduction in False Negatives indicate that the MRFM successfully mitigates the missed detections of tiny aerial targets. The leading IDF1 validates the H-GSTR. By strictly enforcing kinematic smoothness, it effectively suppresses identity switches caused by occlusion and irregular motion. Detailed qualitative comparisons and visual demonstrations of these modules are provided below.

Fig. 5 presents a qualitative comparison between our proposed method and six representative methods, including DeepSORT (Wojke et al., 2017), ByteTrack (Zhang et al., 2022), HybridSORT (M. Yang et al., 2024), ImproAssoc (Stadler and Beyerer, 2023), OCSORT (Cao et al., 2023), and TrackTrack (Shim et al., 2025), on the UAVDT dataset. To ensure a fair comparison, all methods are evaluated on identical frame intervals. The evaluation sequence covers frames 101, 117, 133, and 149. This sequence captures a nighttime traffic scenario characterized by challenging conditions such as low illumination, vehicle interactions, and occlusions caused by vertical structures (e.g., street lamp poles). In the visualization, the dotted lines of distinct colors depict the estimated trajectories of different targets. The “x” markers denote tracking failures—specifically target loss or identity switches (ID Switches)—occurring in the respective frames, while the “✓” symbols signify robust and accurate tracking.

Robustness against occlusion (red and yellow trajectories). We specifically analyze the robustness against occlusion through two representative cases involving blue taxis. Regarding the red trajectory (spanning Frames 1–4), an interruption occurs in Frame 2 when the target is occluded by a lamp post. As indicated by the red “x” markers, this visual gap results in tracking termination for all six comparative methods, which fail to predict the target position behind the obstacle. Concerning the yellow trajectory (spanning Frames 3–4), the TrackTrack algorithm suffers from a missed detection in Frame 3. Furthermore, in Frame 4, when the vehicle is occluded by a street lamp, the remaining comparative methods also experience tracking interruptions, as marked by the yellow “x” symbols. In contrast, our method maintains continuous tracking for both targets throughout these intervals. This stability is attributed to the H-GSTR module, which utilizes curvature-aware manifold restoration to infer latent kinematic states, thereby bridging the occlusion-induced gaps without fragmentation.

Small target perception in low light. Beyond occlusion, the scene highlights the difficulty of tracking small-scale objects in the periphery under poor illumination. As illustrated by the white (spanning Frames 1–4), purple (spanning Frames 1–3), and green trajectories (spanning Frames 1–4) corresponding to TrackTrack (Shim et al., 2025), it suffers from varying degrees of missed detections for distant small vehicles under nighttime conditions. And our method consistently localizes these targets. This validates the efficacy of the MRFM, which adaptively enhances local edge details against background noise via bi-granularity feature decomposition, ensuring high-quality detection inputs when targets occupy minimal pixels.

The visualization confirms that the synergistic integration of MRFM (for precise detection) and H-GSTR (for robust association) enables our network to outperform existing competitive methods, particularly in scenarios involving complex occlusions and scale variations.

4.4.2. Comparison on the VisDrone-MOT dataset

As shown in Table 3, VisDrone-MOT dataset is particularly challenging due to crowded scenes, frequent occlusions, and scale variation. Despite these difficulties, our method achieves the best performance across all key metrics, recording 48.024% HOTA, 44.632% MOTA,

Table 2

Comparison with advanced MOT methods on the UAVDTdataset. The best results are highlighted in **bold** and the second-best results are highlighted in underline format.

Method	Main metrics			Reference metrics				
	HOTA ↑	MOTA ↑	IDF1 ↑	FN ↓	FP ↓	IDs ↓	MT ↑	ML ↓
SORT (Bewley et al., 2016)	60.400	66.414	77.119	19 034	6505	160	201	36
DeepSORT (Wojke et al., 2017)	61.970	68.498	78.615	20 035	4008	61	179	43
FairMOT (Zhang et al., 2021)	49.126	51.189	66.471	33 102	4136	110	107	87
ByteTrack (Zhang et al., 2022)	62.179	68.754	78.760	20 010	3796	102	182	42
ByteTrack+ReID (Zhang et al., 2022)	61.621	68.365	77.205	18 067	6021	118	196	44
BoT-SORT (Aharon et al., 2022)	61.369	67.741	78.508	20 296	4323	64	181	43
UAVMOT (Liu et al., 2022)	61.220	67.901	78.084	19 371	5121	69	190	42
TrackFormer (Meinhardt et al., 2022)	43.165	37.924	53.343	45 197	5585	680	67	75
OCSORT (Cao et al., 2023)	–	47.500	64.900	148 378	47 681	288	–	–
STCMOT (J. Ma et al., 2024)	–	49.200	69.800	99 547	72 901	665	664	203
DroneMOT (P. Wang et al., 2024)	–	50.100	69.600	112 548	57 411	129	638	178
UCMCTrack (Yi et al., 2024)	54.075	61.037	65.860	25 888	2482	984	123	51
SAM2MOT (J. Jiang et al., 2025)	–	66.100	79.300	74 586	40 692	136	816	147
TrackTrack (Shim et al., 2025)	57.662	58.338	71.277	30 483	6617	1152	137	77
DFA-MOT (Zheng et al., 2025)	–	49.900	69.300	119 248	59 218	396	684	230
MOSAIC-Tracker (Zou et al., 2025)	<u>64.568</u>	<u>70.294</u>	<u>81.667</u>	19 584	3137	375	190	38
Ours	65.766	72.314	82.499	19 259	5304	43	200	36

Table 3

Comparison with advanced MOT methods on the VisDrone-MOT dataset. The best results are highlighted in **bold** and the second-best results are highlighted in underline format.

Method	Main metrics			Reference metrics				
	HOTA ↑	MOTA ↑	IDF1 ↑	FN ↓	FP ↓	IDs ↓	MT ↑	ML ↓
SORT (Bewley et al., 2016)	35.080	33.148	42.844	112 481	20 392	3525	329	523
DeepSORT (Wojke et al., 2017)	36.921	34.397	46.714	110 989	21 077	1784	386	517
FairMOT (Zhang et al., 2021)	31.102	12.806	37.745	114 834	59 997	3072	397	581
MOTR (Zeng et al., 2022)	–	22.800	41.400	147 937	28 407	959	272	825
ByteTrack (Zhang et al., 2022)	40.661	39.541	50.398	105 518	16 257	1581	507	538
ByteTrack+ReID (Zhang et al., 2022)	41.422	40.88	51.264	103 245	15 254	1512	510	525
BoT-SORT (Aharon et al., 2022)	42.420	<u>41.652</u>	56.843	103 505	14 114	1430	543	537
UAVMOT (Liu et al., 2022)	38.133	38.685	45.150	108 134	13 610	3357	463	557
TrackFormer (Meinhardt et al., 2022)	35.344	25.000	51.000	141 526	25 856	1534	515	946
OCSORT (Cao et al., 2023)	–	39.600	50.400	123 513	14 631	986	–	–
STCMOT (J. Ma et al., 2024)	–	41.200	52.000	94 445	36 428	3984	667	453
MGTrack (Ren et al., 2024)	–	32.800	44.800	70 346	15 236	1800	–	–
AdaptTrack (K. Li et al., 2024)	42.900	41.300	53.600	106 315	13 965	1426	–	–
UCMCTrack (Yi et al., 2024)	37.072	28.114	38.434	150 590	9244	7213	180	826
TSsm (Zhang and Ye, 2025)	43.892	41.256	54.637	–	–	878	–	–
TrackTrack (Shim et al., 2025)	42.530	37.958	50.956	135 375	51 351	5120	526	656
MOSAIC-Tracker (Zou et al., 2025)	<u>46.044</u>	41.358	<u>58.142</u>	126 112	26 895	1862	583	591
Ours	48.024	44.632	59.668	122 923	25 045	424	637	608

and 59.668% IDF1. Notably, compared to the state-of-the-art method MOSAIC-Tracker (Zou et al., 2025).

The advantage of our method is particularly evident when compared to general-purpose baselines. While BoT-SORT (Aharon et al., 2022) delivers a competitive MOTA (41.652%), it suffers from frequent identity switches (1430 IDs) in crowded scenes. In contrast, our method improves MOTA by nearly 3.0% while dramatically reducing ID switches by over 70.0% (to just 424 IDs). Furthermore, against recent 2025 algorithms like TSsm (Zhang and Ye, 2025) and TrackTrack (Shim et al., 2025), our method maintains a comprehensive lead, proving its adaptability to the scale variations and complex motion patterns inherent in drone surveillance.

The quantitative superiority presented above can be explicitly traced back to the synergistic design of our proposed modules. Specifically, the leading MOTA score of 44.632% validates the effectiveness of the MRFM, which preserves essential feature details. Even more, it achieves exceptional trajectory stability, evidenced by a low number of Identity Switches (424). This stands in striking contrast to top-performing competitors like MOSAIC-Tracker (1862 IDs) and BoT-SORT (1430 IDs). Such a substantial margin provides support for the H-GSTR, demonstrating its capacity to enforce kinematic smoothness and effectively prevent noise-induced fragmentations.

Fig. 6 illustrates a comparative visualization between our method and representative state-of-the-art methods on the VisDrone-MOT benchmark. To ensure a fair comparison, all methods are evaluated on identical frame intervals. The four sampled timestamps (Frames 1–4) correspond to indices 380, 390, 400, and 410, respectively. In the visualization, distinct targets are annotated with colored trajectories, where ‘✓’ denotes successful tracking and ‘x’ indicates failures such as target loss or significant bounding box drift. The results demonstrate that while competing methods frequently succumb to tracking failures in scenarios involving occlusion (e.g., by trees or streetlights), our proposed network exhibits superior robustness under similar conditions. We analyze specific trajectories below to highlight these advantages.

Robustness to appearance inconsistency (red, white, & green trajectories). We first examine the challenges posed by large-scale freight vehicles, identified by the red (Frames 1–4), white (Frames 1–2) and green (Frames 3–4) trajectories. These targets represent a distinct difficulty in the VisDrone-MOT dataset: sparse training samples coupled with significant intra-class variance, such as mismatches between the cab and trailer colors or the presence of canvas covers. As visualized, six competing methods fail to consistently track the red target in Frames 1 and 4, and similarly lose the green target in Frame 4, marked by corresponding ‘x’ symbols. These failures stem from the inability of traditional feature extractors to generalize across such heterogeneous

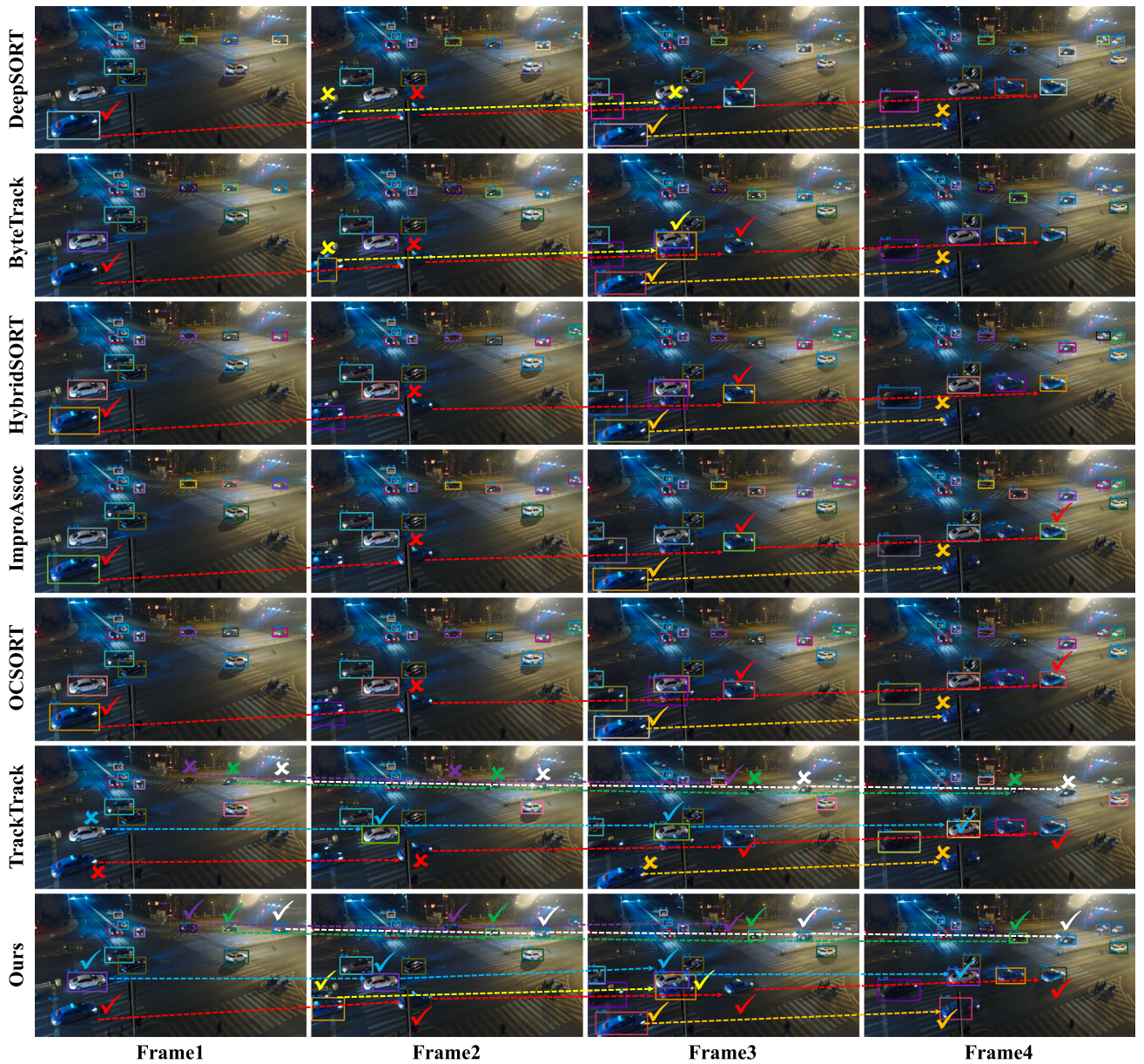


Fig. 5. Visual comparison of the proposed method against six benchmarks under a nighttime scenario from the UAVDT dataset. This sequence features challenges including low light, dense traffic, and occlusions (street poles). Colored dotted lines represent the tracking trajectories. The symbols “✗” and “✓” indicate tracking errors (loss or ID switches) and successful tracking, respectively.

appearances. Regarding the white trajectory, all six comparative algorithms experience tracking loss for a truck in Frame 2, as indicated by the white ‘✗’ symbol. In contrast, our method maintains stable tracking for all trucks. This is largely attributed to the MRFM module, which adaptively modulates the receptive field to capture both the local details of the vehicle cab and the global context of the trailer, ensuring robust feature representation despite appearance variations.

Small target perception and occlusion (yellow, orange, & blue trajectories). Parallel challenges are observed in tracking small-scale targets. Specifically, the yellow (car) and orange (tricycle) trajectories span Frames 1–3. In this interval, the targets appear at a small image scale, yielding weak and ambiguous appearance cues; consequently, all five compared trackers terminate the tracks, as indicated by the yellow and orange ‘✗’ markers. Similarly, the blue trajectory (Frames 1–4) corresponds to a far-field van that appears small in the image and is partially occluded by a foreground vehicle. While the TrackTrack algorithm fails to maintain the bounding box (the blue ‘✗’ symbol) due

to the target’s small scale and weak visual cues, our method successfully preserves the trajectory.

4.5. Ablation study

4.5.1. The ablation of components

To evaluate the individual efficacy and the contribution of the proposed components, we conduct comprehensive ablation studies on both the UAVDT and VisDrone-MOT benchmarks. We adopt the BoT-SORT (Aharon et al., 2022), as our baseline. Based on this, we integrate the MRFM and the H-GSTR strategy. The quantitative comparisons are detailed in Table 4.

Effectiveness of MRFM. As shown in the second row of Table 4, incorporating the MRFM module into the baseline yields consistent performance gains. On the challenging VisDrone-MOT dataset, which is dominated by small-scale targets, MRFM significantly boosts the MOTA from 38.413% to 41.492% (an improvement of +3.08%). This

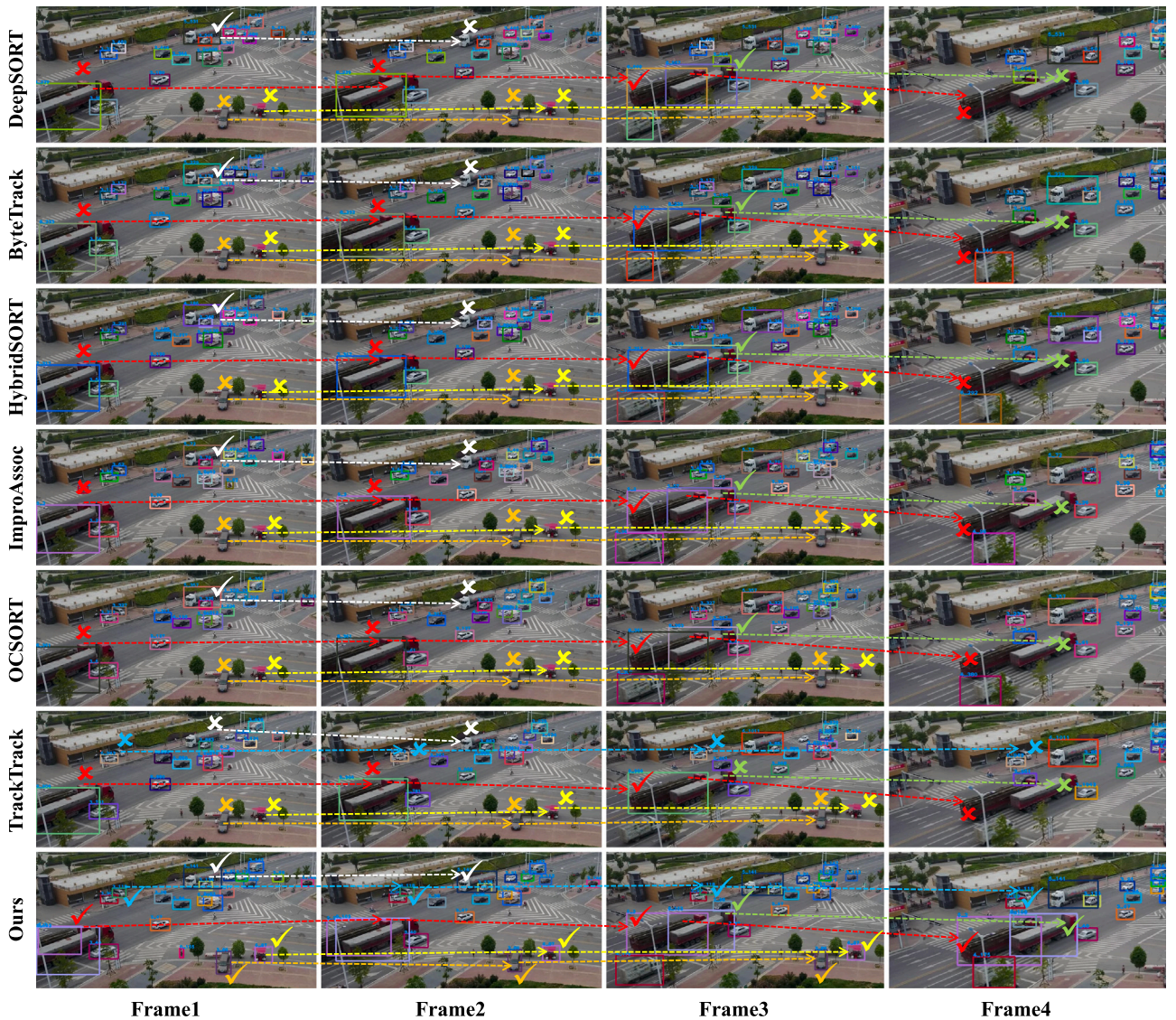


Fig. 6. Visual comparison of the proposed method against six benchmarks under a nighttime scenario from the VisDrone-MOT dataset. This sequence features challenges including low light, dense traffic, and occlusions (street poles). Colored dotted lines represent the tracking trajectories. The symbols ‘x’ and ‘✓’ indicate tracking errors (loss or ID switches) and successful tracking, respectively.

Table 4
Ablation study of different modules on UAVDT and VisDrone-MOT.

Module	UAVDT			VisDrone-MOT		
	HOTA↑	MOTA↑	IDF1↑	HOTA↑	MOTA↑	IDF1↑
Baseline	61.551	67.140	78.647	44.128	38.413	55.131
Only MRFM	63.338	68.831	80.488	45.468	41.492	56.816
Only H-GSTR	64.133	70.628	80.714	47.083	41.573	58.613
MRFM & H-GSTR	65.766	72.314	82.499	48.024	44.632	59.668

substantial increase in detection accuracy validates our hypothesis that the bi-granularity feature decomposition effectively addresses the scale-resolution dilemma, enabling the tracker to perceive tiny objects that are otherwise missed by fixed-receptive-field backbones.

Effectiveness of H-GSTR. The third row demonstrates the impact of the H-GSTR module. Independent of feature enhancements, utilizing H-GSTR alone improves the HOTA on UAVDT by +2.58% (from 61.551% to 64.133%) and on VisDrone-MOT by +2.96%. More importantly, the IDF1 scores see a steady rise across both datasets. This indicates that the curvature-aware manifold restoration successfully

bridges fragmented trajectories caused by occlusion, thereby enhancing the temporal consistency and smoothness of the tracking results.

Synergy of the full network. Finally, the last row presents the results of the full method with both MRFM and H-GSTR. It achieves the best performance across all metrics, surpassing the baseline by a remarkable margin of +4.22% HOTA on UAVDT and +6.22% MOTA on VisDrone-MOT. The results confirm that they are highly complementary: MRFM provides high-quality detection candidates to reduce false negatives, while H-GSTR ensures these candidates are associated into physically plausible trajectories.

4.5.2. Detector-side ablation of MRFM

Since MRFM is introduced on the detector side, we further evaluate its contribution using detector-level metrics on both VisDrone and UAVDT. Under the same detector backbone and training/testing protocol, we compare MRFM with representative detector-side alternatives, including DynamicConv, Restormer, and AMFE, and report Recall (R), mAP50, and mAP50-95. The results are summarized in Table 5. The detector-side results reveal a clear and meaningful trend. Across both datasets, MRFM delivers the best overall detection performance, indicating that its effectiveness arises already at the feature-extraction

Table 5

Detector-side ablation of MRFM on VisDrone and UAVDT. The proposed MRFM achieves the best overall detection performance on both datasets, validating that its gains originate not only from downstream association but also from stronger feature extraction at the detection stage.

Method	VisDrone			UAVDT		
	R↑	mAP50↑	mAP50-95↑	R↑	mAP50↑	mAP50-95↑
DynamicConv (Han et al., 2024)	0.342	0.327	0.170	0.732	0.796	0.482
Restormer (H. Li et al., 2025)	0.366	0.359	0.187	0.735	0.769	0.448
AMFE (Zou et al., 2025)	0.389	0.364	0.175	0.786	0.839	0.517
Ours (MRFM)	0.389	0.369	0.194	0.793	0.845	0.518

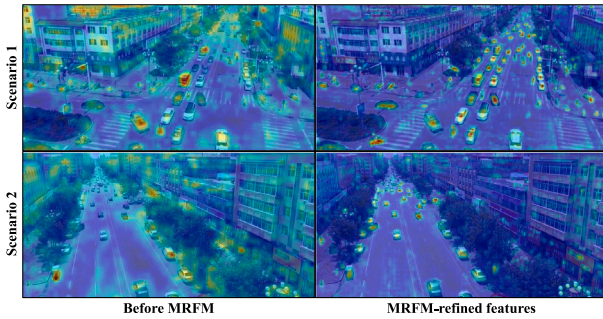


Fig. 7. Qualitative comparison of feature saliency maps before MRFM and after MRFM-refined feature propagation in two representative aerial traffic scenes.

stage rather than being introduced solely through downstream trajectory association. More importantly, the improvement is particularly pronounced on the more stringent mAP50-95 metric, especially on VisDrone. This finding suggests that MRFM does not merely enhance coarse object responses, but more importantly improves fine-grained localization accuracy under challenging conditions such as dense clutter, severe scale variation, and tiny-object scenarios.

Notably, the detector-side gains brought by MRFM extend beyond a simple improvement in recall. On both VisDrone and UAVDT, its consistent advantage on mAP50-95 demonstrates superior localization quality under stricter overlap criteria. This is particularly important for aerial multi-object tracking, where even minor localization errors can substantially undermine downstream data association and trajectory continuity. Therefore, the observed trend indicates that MRFM improves not only the detectability of small targets, but also the precision with which they are localized in cluttered aerial scenes.

From a mechanistic perspective, these results are well aligned with the design of MRFM. Specifically, the local-detail branch preserves weak fine-grained target cues that are otherwise easily suppressed by aggressive downsampling, whereas the global-context branch provides broader scene awareness in the presence of viewpoint variation and complex background interference. The context-gated modulation and structure-preserving channel interaction further strengthen target-relevant responses while maintaining spatial topology. As a result, the detector-side improvements reported in Table 5 should not be viewed as merely empirical performance gains; rather, they provide evidence that MRFM fulfills its intended functional role in aerial small-target perception.

To further illustrate the effect of MRFM on small-target representation, we provide a qualitative comparison of feature saliency maps before MRFM and after MRFM-refined feature propagation in two representative aerial traffic scenes, as shown in Fig. 7. Scenario 1 corresponds to a dense urban intersection scene, where multiple small vehicles appear together with cluttered background structures such as buildings, road markings, and roadside facilities. Before MRFM, the feature responses are relatively diffuse and are distributed over both target regions and surrounding background structures. After MRFM refinement, the responses become more concentrated on small vehicles and

the traffic-flow region, with target-related activations more clearly organized along the road direction. This indicates that MRFM strengthens the separation between target-relevant cues and surrounding structural clutter in dense intersection scenes.

In Scenario 2, a long road corridor affected by interference from roadside buildings and trees, the pre-MRFM features still show broad background responses, indicating limited emphasis on truly informative small-target regions. By contrast, the MRFM-refined features exhibit more compact and target-oriented activations over small vehicles and adjacent informative road regions. Meanwhile, the responses along the vehicle distribution become more structured, whereas irrelevant background activations are relatively suppressed. This is particularly important for aerial tracking, where distant vehicles often provide weak appearance evidence that is easily corrupted by surrounding textures.

Overall, these visual results indicate that MRFM does not merely increase response magnitude globally, but instead enhances the concentration and discriminability of small-target features in cluttered aerial scenes. This behavior is consistent with the design of MRFM, in which the local-detail branch preserves weak fine-grained cues, the global-context branch captures broader contextual dependencies, and the context-gated modulation adaptively strengthens target-relevant responses. Hence, the qualitative saliency analysis complements the quantitative detection and tracking results by showing how MRFM reorganizes feature representations toward more target-oriented patterns.

4.5.3. Comparative advantages of MRFM

To further substantiate the superiority of MRFM, we benchmark it against representative dynamic convolution-based modules, including AMFE from the recent MOSAIC-Tracker (Zou et al., 2025), DynamicConv (Han et al., 2024), and CMUNeXtB (Tang et al., 2024).

As reported in Table 6, MRFM consistently achieves the best performance across all three MOT metrics on both VisDrone-MOT and UAVDT, indicating that its gains are not limited to detection accuracy but also translate to stronger identity preservation. On VisDrone-MOT, MRFM surpasses the strongest competitor AMFE (Zou et al., 2025) by +2.38% HOTA, +3.11% MOTA, and +2.04% IDF1, showing a clear advantage under more severe scale variation and clutter. On UAVDT, MRFM maintains a stable lead over AMFE, suggesting that the proposed design generalizes across different aerial domains.

The observed improvements can be attributed to the fact that generic dynamic convolution modules (e.g., DynamicConv (Han et al., 2024) and CMUNeXtB (Tang et al., 2024)) primarily rely on dynamically generated kernels or routing decisions, but do not explicitly address the feature submergence phenomenon that is prevalent in aerial imagery, where tiny targets are easily dominated by background textures and abrupt scale swings. In contrast to generic dynamic convolution modules, MRFM is tailored to aerial tracking by explicitly decoupling features into complementary granularities: a local-detail branch that preserves fine-grained boundary cues for tiny targets, and a global-context branch that stabilizes broader contextual semantics under viewpoint changes. Meanwhile, the structure-preserving channel mixer strengthens discriminability without collapsing spatial topology, which is particularly beneficial when targets are heavily down-scaled, partially occluded, or embedded in cluttered backgrounds. Together, these designs provide stronger detection-quality signals and more reliable appearance cues, thereby improving both association robustness and overall tracking performance.

Table 6

Comparison of MRFM with representative dynamic convolution-based modules on VisDrone-MOT and UAVDT.

Method	VisDrone-MOT			UAVDT		
	HOTA↑	MOTA↑	IDF1↑	HOTA↑	MOTA↑	IDF1↑
AMFE (Zou et al., 2025)	43.091	38.381	54.777	63.273	68.641	80.200
DynamicConv (Han et al., 2024)	42.210	31.240	52.314	58.352	61.263	74.284
CMUNeXtB (Tang et al., 2024)	40.726	29.615	49.958	59.534	60.074	75.363
MRFM (Ours)	45.468	41.492	56.816	63.338	68.831	80.488

Table 7

Ablation study of the ReID branch and the proposed H-GSTR module on UAVDT and VisDrone. Best results for each metric are shown in bold.

Dataset	Baseline	ReID	H-GSTR	HOTA ↑	MOTA ↑	IDF1 ↑
UAVDT	BoT-SORT	×	×	63.338	68.831	80.488
	BoT-SORT	✓	×	63.327	68.827	80.471
	BoT-SORT	×	✓	65.760	72.309	82.497
	BoT-SORT	✓	✓	65.739	72.322	82.467
VisDrone	BoT-SORT	×	×	45.431	41.497	56.748
	BoT-SORT	✓	×	45.468	41.492	56.816
	BoT-SORT	×	✓	47.976	44.602	59.612
	BoT-SORT	✓	✓	48.028	44.605	59.672

Table 8

Comparison of interpolation schemes used in TCR under a unified post-processing setting. The proposed PCHIP-based H-GSTR achieves the best results on both UAVDT and VisDrone.

Method	UAVDT			VisDrone		
	HOTA↑	MOTA↑	IDF1↑	HOTA↑	MOTA↑	IDF1↑
4th-order polynomial interpolation	53.714	46.795	68.586	34.488	11.751	40.715
5th-order polynomial interpolation	57.413	55.658	73.472	36.220	15.983	43.215
Ours	65.751	72.309	82.496	48.029	44.697	59.668

4.5.4. Ablation study of the ReID branch

To further clarify the role of the ReID branch in the BoT-SORT baseline, we additionally evaluate four settings on both UAVDT and VisDrone: BoT-SORT, BoT-SORT + ReID, BoT-SORT + H-GSTR, and BoT-SORT + ReID + H-GSTR. During inference, the ReID branch is enabled by the `--reid` option in the tested configuration. As shown in Table 7, the tested ReID branch contributes limited additional benefit under the current setting. On UAVDT, enabling ReID leads to only negligible variation relative to the baseline, suggesting that the tested appearance cue provides little extra discrimination in this dataset. By contrast, introducing H-GSTR yields clear and stable gains across all three metrics, indicating that the main improvement comes from trajectory refinement rather than from the tested ReID association branch. On VisDrone, ReID provides a small positive gain, but its contribution remains noticeably smaller than that of H-GSTR. This difference is meaningful: compared with the tested ReID configuration, H-GSTR brings much more substantial improvements by directly repairing temporal fragmentation and enhancing trajectory continuity, which is especially beneficial in complex aerial scenes. Overall, these results suggest that, under the current implementation and tested model/domain setting, H-GSTR is the dominant source of performance improvement. We note that the effectiveness of ReID may depend on the selected model architecture and checkpoint, especially under cross-domain conditions. Exploring stronger or more domain-specific ReID models remains a meaningful direction for future work.

We emphasize that this observation does not imply that ReID is generally ineffective. Rather, its practical contribution may depend on the selected model architecture, checkpoint, and domain adaptation quality, especially in aerial tracking scenarios with small objects and weak appearance cues. Exploring stronger or more domain-specific ReID models remains an interesting direction for future work.

4.5.5. Interpolation method comparison for TCR

To further justify the adoption of PCHIP in H-GSTR, we compare it with representative higher-order polynomial interpolation schemes under a unified post-processing setting on UAVDT and VisDrone. Specifically, we evaluate 4th-order polynomial interpolation, 5th-order polynomial interpolation, and our PCHIP-based formulation while keeping the rest of the tracking pipeline unchanged. As shown in Table 8, our method achieves the best performance on both datasets.

These results suggest that the effectiveness of H-GSTR does not come from using a higher interpolation order, but from adopting a more suitable recovery bias for bounded trajectory refinement. In aerial tracking, especially for tiny targets, overly flexible polynomial fitting is more likely to introduce overshoot and oscillatory trajectory evolution, which can degrade both localization quality and downstream association stability. By contrast, the proposed PCHIP-based TCR preserves local geometric continuity and monotonic trends, making it better aligned with the short-gap recovery objective of H-GSTR. It is notable that the same pattern is consistently observed on both UAVDT and VisDrone, despite their different scene characteristics and target distributions. This indicates that the advantage of PCHIP is not benchmark-specific, but reflects a more general suitability for aerial trajectory recovery, where local plausibility and geometric stability are more important than aggressive fitting flexibility.

4.5.6. Visualization analysis of H-GSTR

To intuitively verify the efficacy of the High-Order Geometric Stabilized Trajectory Refinement (H-GSTR) strategy, we conduct a qualitative ablation study. We present visual comparisons between the baseline (w/o GSTR) and our full model (w/ GSTR) separately for the UAVDT and VisDrone-MOT benchmarks. In the visualization, distinct targets within the same scenario are differentiated by colored trajectories, where ‘✓’ denotes successful tracking and ‘×’ indicates failure (e.g., target loss or drift). Based on this setup, we analyze representative scenarios to elucidate the specific contributions of the TCR and KNS sub-modules.

Analysis on the UAVDT dataset. Fig. 8 illustrates the performance comparison on the UAVDT benchmark, focusing on challenges related to structural occlusion and illumination variation. First, in Scenario 2 (middle row), the target vehicle undergoes an occlusion by a street lamp post. While the baseline tracker succumbs to this topological break—evidenced by the immediate target loss in Frame 2—our method maintains a continuous trajectory. This success explicitly validates the TCR sub-module, which utilizes monotonicity-constrained interpolation to physically reconstruct the latent motion path during visual gaps. Second, Scenario 3 (bottom row) depicts a challenging low-light environment where a blue taxi is partially occluded by an electric pole. The weak visual cues combined with occlusion cause the baseline bounding box to exhibit severe drift and eventual termination. In contrast, our method produces a smooth and stable trajectory. This robustness confirms the synergy of our sub-modules: KNS suppresses the observation noise induced by low-light conditions, while TCR bridges the interruption caused by the pole, ensuring consistent tracking under adverse lighting.

Analysis on the VisDrone-MOT dataset. Complementary evidence is provided in the VisDrone-MOT dataset comparisons shown in Fig. 9. In Scenario 1 (top row), the vehicle traverses behind roadside vegetation, leading to partial occlusion. The baseline tracker fails to predict

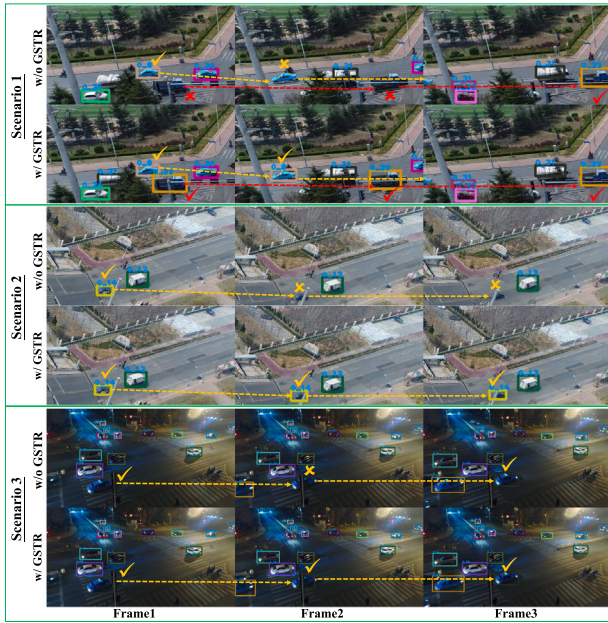


Fig. 8. Qualitative ablation on the UAVDT dataset. We benchmark the baseline (w/o H-GSTR) against the proposed network (w/ H-GSTR) in challenging aerial scenarios. Colored dashed lines represent the estimated trajectories, where ✓ and × indicate successful tracking and failure (e.g., target loss or drift), respectively.

Table 9

Sequence-specific analysis on the pedestrian-heavy VisDrone sequence uav0000088_00290_v.

Method	HOTA ↑	MOTA ↑	IDF1 ↑
w/o H-GSTR	31.566	33.098	40.874
w/ H-GSTR	34.222	37.286	43.712

the non-linear motion correctly during this interval, resulting in eventual tracking termination. Conversely, our method robustly recovers the trajectory, further demonstrating the capability of the TCR sub-module to handle occlusion dynamics in aerial views. Additionally, we examine Scenario 2 (middle) and Scenario 3 (bottom), which highlight the challenges of tracking small-scale vehicles amidst occlusion and clutter. While the baseline suffers from significant drift or loss due to frequent obstructions, our method maintains tight alignment by enforcing geometric smoothness via KNS and recovering missing states via TCR. These results empirically confirm H-GSTR's robustness against the dual challenges of small-target instability and environmental obstruction.

Therefore, the effectiveness of H-GSTR could not be understood as the isolated benefit of classical monotonic interpolation or local polynomial smoothing. Rather, it arises from their structured coupling under an aerial MOT-specific restoration objective, where fragmented topology and jitter-induced instability are addressed jointly to improve geometric continuity, motion consistency, and downstream tracking robustness. Beyond the individual cases above, an important common pattern can be observed in both datasets: the benefit of H-GSTR is most evident when the tracker simultaneously faces incomplete observations and unstable short-term geometry. In such situations, interpolation alone is insufficient because it cannot suppress local jitter, whereas smoothing alone is also insufficient because it cannot restore broken temporal support. The visual results consistently show that H-GSTR improves trajectory continuity without introducing obvious temporal lag or excessive geometric distortion, which supports the necessity of jointly coupling TCR and KNS in a unified refinement framework.

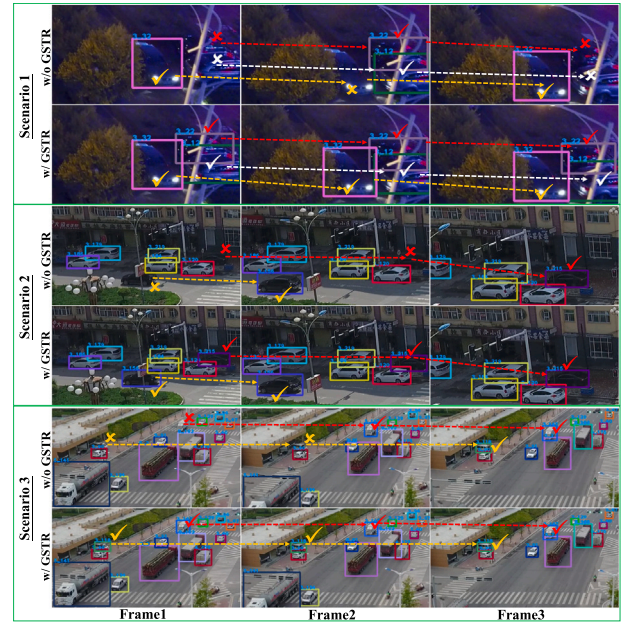


Fig. 9. Qualitative ablation results on the VisDrone-MOT dataset. The comparison demonstrates the robustness of the proposed H-GSTR module against the baseline. The symbols ✓ and × denote the correctness of the tracking results.

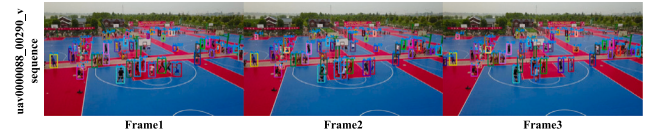


Fig. 10. Qualitative analysis on a pedestrian-heavy VisDrone sequence uav0000088_00290_v, with a crowded basketball-court scene with interactions and highly non-linear pedestrian motion.

4.5.7. Analysis on a pedestrian-heavy non-linear scene

To examine the robustness of H-GSTR under highly non-linear motion, we conduct a sequence-specific analysis on the pedestrian-heavy VisDrone sequence uav0000088_00290_v. This sequence depicts a crowded basketball-court scene with frequent pedestrian interactions, abrupt motion changes, and strong local non-linearity, and therefore constitutes a substantially more stringent test than vehicle-dominated aerial sequences. Since pedestrian motion in such scenarios is often irregular and highly dynamic, this sequence provides an effective test bed for examining whether the proposed bounded interpolation and local regularization strategy may cause oversmoothing or trajectory drift.

As shown in Table 9, applying H-GSTR improves HOTA from 31.566% to 34.222%, MOTA from 33.098% to 37.286%, and IDF1 from 40.874% to 43.712% on this sequence. These consistent gains are particularly meaningful because they are obtained on a scene dominated by pedestrians with frequent interactions and highly non-linear local motion. If the proposed refinement were prone to excessive smoothing, one would expect the benefits to diminish or even reverse under such conditions. Instead, the sequence-level improvements indicate that the current bounded interpolation and local regularization strategy remains practically robust and does not introduce severe trajectory drift in this challenging scenario.

We further provide qualitative visualizations at fixed frame intervals, as shown in Fig. 10. The selected frames illustrate a crowded basketball-court scene in which pedestrians exhibit abrupt motion changes, close interactions, and intermittent partial occlusions. Although some missed detections still occur in this intrinsically difficult

Table 10

Cross-tracker generalization study of H-GSTR on UAVDT. H-GSTR consistently improves tracking performance across three recent MOT trackers, indicating that its effectiveness is not limited to the BoT-SORT framework.

Method	H-GSTR	HOTA ↑	MOTA ↑	IDF1 ↑
TrackTrack (Shim et al., 2025)	×	53.112	49.575	66.633
	✓	58.881	65.385	74.807
HybridSORT (M. Yang et al., 2024)	×	61.369	68.293	76.451
	✓	62.327	69.907	76.819
UCMCTrack (Yi et al., 2024)	×	57.522	65.002	74.132
	✓	58.881	65.385	74.807

sequence, no evident oversmoothing-induced drift or systematic trajectory distortion is observed after applying H-GSTR. Instead, the refined trajectories remain visually stable and preserve plausible local motion behavior, which is consistent with the quantitative gains reported above.

This sequence-specific result is important because it complements the dataset-level averages with a more targeted stress test. Compared with vehicle-dominated scenes, crowded pedestrian motion contains more abrupt direction changes, closer inter-object interaction, and weaker short-term regularity. Under such conditions, a refinement strategy that is overly aggressive would be more likely to produce motion drift or temporal misalignment. The fact that H-GSTR still improves all three metrics while preserving visually plausible local motion suggests that the proposed refinement remains sufficiently conservative and does not rely on overly restrictive linear-motion assumptions.

4.6. Cross-tracker generalization of H-GSTR

To further verify that H-GSTR is not limited to the BoT-SORT framework, we additionally integrate it into three recent MOT trackers on UAVDT, namely TrackTrack (Shim et al., 2025), HybridSORT (M. Yang et al., 2024), and UCMCTrack (Yi et al., 2024). The corresponding results are reported in Table 10. As shown in the table, introducing H-GSTR consistently improves tracking performance across all three frameworks, which indicates that its effectiveness is not tied to a specific tracker architecture or association strategy. Instead, its benefit mainly arises from trajectory-level refinement on associated track hypotheses, supporting its use as a generally applicable post-processing module beyond the BoT-SORT baseline.

The relative gains also reveal a meaningful trend. The improvement is most pronounced on TrackTrack, where HOTA increases from 53.112% to 58.881% and MOTA rises from 49.575% to 65.385%. This suggests that H-GSTR is particularly effective when the upstream trajectories still contain noticeable short-term fragmentation or local geometric instability, because bounded recovery and local stabilization can then provide substantial refinement benefits. By contrast, the gain on HybridSORT is more moderate, indicating that when the baseline trajectory quality is already relatively strong, H-GSTR mainly serves as a stable refinement module rather than a large corrective module.

UCMCTrack Tracker provides a further complementary case. Although it already incorporates camera-motion-aware modeling, adding H-GSTR still yields consistent gains, showing that the proposed refinement is not limited to trackers without motion compensation. This observation suggests that H-GSTR complements, rather than replaces, upstream tracking mechanisms: it improves the completeness and local consistency of the resulting trajectories, while the extent of improvement naturally depends on the baseline stability of the upstream tracker. Overall, these cross-tracker results provide additional evidence that H-GSTR has broader applicability as a tracker-agnostic post-processing trajectory refinement module.

Table 11

Efficiency analysis comparing inference speed (FPS) and tracking performance against state-of-the-art methods on the UAVDT benchmark. The best trade-off results are highlighted.

Method	HOTA↑	MOTA↑	IDF1↑	FPS↑
MM-Tracker (Yao et al., 2025)	–	51.400	68.900	31.100
UCMCTrack (Yi et al., 2024)	54.075	61.037	65.860	15.430
MOSAIC (Zou et al., 2025)	64.568	70.294	81.667	14.094
Ours	65.766	72.314	82.499	21.620

4.6.1. The speed ablation

To evaluate the practical deployability of the proposed network, we analyze the trade-off between inference speed (FPS) and tracking performance on UAVDT. The results, evaluated on a single NVIDIA RTX A6000 GPU, are summarized in Table 11.

As evidenced in Table 11, our method achieves the most optimal balance between accuracy and efficiency. Compared to the top-performing competitor MOSAIC-Tracker (Zou et al., 2025), our network not only surpasses it in tracking accuracy (+1.20% HOTA and +2.02% MOTA) but also delivers a substantial improvement in processing speed, increasing from 14.094 FPS to 21.620 FPS (+53.44%). This acceleration implies that our MRFM and H-GSTR modules are computationally efficient despite their complex designs. While MM-Tracker (Yao et al., 2025) achieves a higher frame rate (31.100 FPS), it comes at the cost of significant performance degradation, lagging behind our method by over 20.91 points in MOTA. In contrast, our method maintains a running speed of 21.620 FPS, which is sufficient for near-real-time applications on standard hardware, without compromising state-of-the-art tracking precision.

4.7. Quantitative analysis of trajectory recovery

To quantify the contribution of the H-GSTR module in resolving trajectory fragmentation, we conduct a sequence-level recovery analysis on the UAVDT benchmark. We define the Recovery Rate (R_{rec}) as the percentage of valid detections recovered by our module relative to the raw detections generated by the baseline tracker: $R_{rec} = (N_{refined} - N_{raw}) / N_{raw} \times 100\%$.

Table 12 details the statistical results across ten diverse scenarios, sorted by recovery efficacy. The results reveal a strong correlation between scene complexity and the recovery capability of H-GSTR. The visual comparison of the detection increments is presented in Fig. 11. As observed, the H-GSTR module provides the most significant boost in scenarios characterized by occlusion and ego-motion.

Analysis of challenges. The most significant improvement is observed in sequence **M0601** (High Viewpoint/Small Targets), achieving a remarkable recovery rate of **27.27%**. In this scenario, targets appear small due to the high flight altitude, causing the baseline detector to frequently miss objects during rapid motion. H-GSTR effectively bridges these long-term detection gaps by exploiting the temporal continuity of the target manifold, recovering 2841 valid bounding boxes that were topologically broken. Similarly, in **M1202** (10.35%), which features frequent occlusions by trees, our TCR component of H-GSTR successfully hallucinates the latent trajectory segments hidden behind obstacles without generating overshoots.

Robustness in low-illumination conditions. For night-time sequences such as **M1003** (Night/Fast Motion) and **M1008** (Night/Inter section), the module maintains a robust recovery rate of roughly 9.00%. Under low-light conditions, motion blur often degrades feature distinctiveness, leading to fragmented tracklets. The KNS component of H-GSTR mitigates this by smoothing out the stochastic observation noise, thereby preserving trajectory coherence even when visual cues are unreliable.

Conservative refinement in stable scenes. Conversely, in simpler scenarios like **M0101** (1.89%) and **M0203** (3.30%), the recovery rate

Table 12

Quantitative Analysis of Trajectory Recovery on UAVDT Dataset. The sequences are sorted by Recovery Rate in descending order. “Recovered Δ ” denotes the net increase in valid detections after applying H-GSTR.

Sequence	Scenario description	# Dets (Raw)	# Dets (Refined)	Recovered Δ	Recovery rate
M0601	High Viewpoint/Small Targets	9712	12,360	+2,648	27.27%
M1202	Small Targets/Tree Occlusion	4512	4979	+467	10.35%
M1003	Night/Fast Motion	9948	10,868	+920	9.25%
M1008	Night/Intersection	2070	2258	+188	9.08%
M0402	Crowded Intersection	9067	9871	+804	8.87%
M0604	Small Targets	11,560	12,559	+999	8.64%
M1002	Night/Moderate Density	6706	7009	+303	4.52%
M0202	Occlusion/Vehicle Interaction	3457	3574	+117	3.38%
M0203	Low Illumination	5492	5673	+181	3.30%
M0101	Intersection	5398	5500	+102	1.89%
Average	-	6792	7465	+673	8.66%

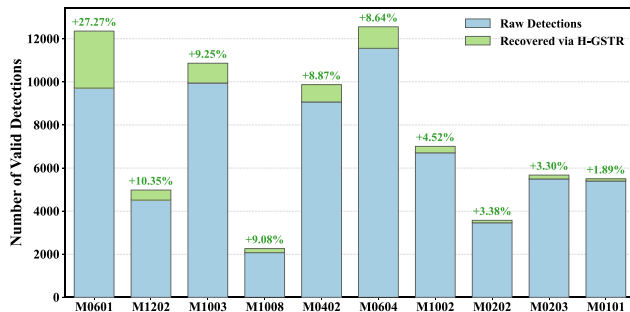


Fig. 11. Quantitative analysis of trajectory recovery on the UAVDT benchmark. The stacked bars illustrate the contribution of the H-GSTR module across ten diverse scenarios. The **light blue** bars represent detections from the baseline tracker, while the **light green** segments denote the valid targets recovered by our geometric refinement strategy. The percentages atop the bars highlight the relative recovery rate (R_{rec}), demonstrating improvements in challenging sequences like M0601 (high viewpoint/small targets) and M1202 (small targets/tree occlusion).

remains modest. This indicates that H-GSTR operates conservatively: it avoids introducing false positives (hallucinations) when the baseline tracker already provides consistent trajectories. This selective activation mechanism validates the robustness of our approach, ensuring that refinement is only applied when geometric inconsistencies are statistically significant.

5. Conclusion

In this paper, we presented a robust network tailored for the challenging task of multi-object tracking in aerial scenarios. By identifying the intrinsic limitations of generic trackers in handling scale variations and irregular motion, we introduced two synergistic components: the MRFM and H-GSTR. MRFM successfully decouples local fine-grained details from global contexts to recover tiny targets, while H-GSTR leverages curvature-aware manifold restoration to bridge occlusion-induced trajectory gaps. Quantitative and qualitative evaluations on the VisDrone-MOT and UAVDT benchmarks demonstrate that our approach not only sets a new state-of-the-art in detection accuracy (MOTA) but also achieves exceptional stability in trajectory continuity (IDF1 and IDs). The visualization further confirms that our model effectively decouples the complexities of small-target perception and occlusion-induced fragmentation.

CRedit authorship contribution statement

Jian Zou: Writing – review & editing, Writing – original draft, Methodology, Investigation. **Ke Liu:** Visualization, Validation, Methodology. **Wei Zhang:** Validation, Supervision, Methodology, Investigation. **Qiang Li:** Supervision, Methodology, Investigation. **Qi Wang:** Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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